HOSPITAL PATIENT REGISTRATION AND APPOINTMENT SYSTEM

AN INDIVIDUAL PROJECT REPORT

Submitted by

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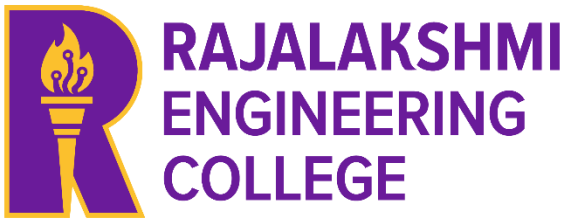
in partial fulfilment of the End Semester Examination for the Open Elective course

OCB2331 – NO-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

MECHANICAL ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "**HOSPITAL PATIENT REGISTRATION AND APPOINTMENT SYSTEM**" is the bonafide work of **ANISHA P** (241101009), Bachelor of Engineering – Mechanical Engineering, who carried out the individual project work on the Zoho Creator no-code platform for the Open Elective course OCB2331 – No-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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INTERNAL EXAMINER	**EXTERNAL EXAMINER**

DECLARATION

I, **ANISHA P** (241101009), hereby declare that the project report titled "**HOSPITAL PATIENT REGISTRATION AND APPOINTMENT SYSTEM**", submitted for the End Semester Examination of the Open Elective course OCB2331 – No-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its forms, reports, pages, rules and workflows — was designed and configured by me using the no-code features of the platform.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder) or external resource used in this work has been duly acknowledged in the report.

Place: Chennai.
Date: 31 October 2026.

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ANISHA P

ABSTRACT

In many small and medium hospitals the first contact with the patient — registration and the booking of an out-patient consultation — is still handled on paper forms, bound registers, diaries and spreadsheets. This leads to duplicate and illegible patient records, double booking of doctor slots, long queues, forgotten appointments, delays in approving doctors' leave and billing discounts, and no timely summary for the administration. The same resource-scheduling and record-keeping problem is met by mechanical engineers in maintenance workshops and service centres.

This project presents the Hospital Patient Registration and Appointment System, built on the Zoho Creator no-code platform without writing any code, so that a non-software engineer can deliver a complete application in a short time. The application contains eight forms (Department, Doctor, Patient Registration, Appointment, Consultation Record, Billing, Doctor Leave Request and Patient Feedback) using 24 field types, lookups, a prescription subform, formula fields, field rules and validations. Ten reports of list, Kanban, calendar, chart, pivot and spreadsheet types and two pages, including an administrator dashboard, present the data. Eight workflows built in the visual workflow builder — four form workflows, two approval processes and two schedules — validate slots, send confirmations and reminders, close appointments, generate bills, route leave and discount requests for approval and mark no-shows. Four user roles with record-level rules protect the data.

The application was tested with eighteen test cases and three users, published to the web, the patient portal and the Zoho Creator mobile app, and compared with the manual process. In trial runs the time to register and book a patient fell from about ten minutes to about three, paper steps were eliminated, and the daily summary became instantly available on the dashboard.

Keywords: *No-Code Development; Zoho Creator; Citizen Developer; Workflow Automation; Dashboard; Hospital Appointment System*

PROJECT AT A GLANCE

Attribute	Details
Project Title	Hospital Patient Registration and Appointment System
Engineering Domain	Mechanical Engineering – Industrial / Service Operations: Resource (Doctor Slot) Scheduling and Record Keeping in a Hospital OPD
Student Name & Register No.	ANISHA P – 241101009
Branch & Semester	Mechanical Engineering – Semester V
Zoho Creator Edition	Education Edition
Application Name	Hospital Patient Registration and Appointment System
Published / Live URL	https://creatorapp.zoho.in/anishap/hospital-patient-registration-and-appointment-system
Demo Credentials (per role)	Administrator: admin.demo@example.com | Receptionist: reception.demo@example.com | Doctor: doctor.demo@example.com | Patient: patient.demo@example.com
Built from	Scratch (AI app builder used only for initial form suggestions)
Number of Forms / Subforms	8 / 1 (Prescription subform)
Number of Reports (by type)	List: 5 | Kanban: 1 | Calendar: 1 | Pivot/Chart: 2 | Others: 1 (Spreadsheet) – Total 10
Number of Pages / Dashboards	2 (Admin Dashboard, Reception Desk)
Number of Workflows	Form: 4 | Schedule: 2 | Approval: 2 | Blueprint: 0 – Total 8
Rules & Formula Fields	12 (4 formula fields, 4 field rules, 1 lookup filter, 3 validation rules)
Roles Defined	Administrator, Receptionist, Doctor, Patient
Mobile App / Portal Used	Yes – Zoho Creator mobile app and patient portal
Date of Publishing	28 October 2026

TABLE OF CONTENTS

ABSTRACTiv

PROJECT AT A GLANCEv

LIST OF TABLESviii

LIST OF FIGURESx

LIST OF ABBREVIATIONSxi

INTRODUCTION1

1.1 Background1

1.2 Problem Statement1

1.3 Objectives2

1.4 Scope of the Project2

1.5 Why No-Code?3

1.6 Organisation of the Report3

LITERATURE SURVEY AND SYSTEM STUDY4

2.1 Study of the Existing System4

2.2 Limitations of the Existing System4

2.3 No-Code, Low-Code and Traditional Development5

2.4 Literature Survey5

2.5 Proposed System6

REQUIREMENT ANALYSIS7

3.1 Users and Roles7

3.2 Functional Requirements7

3.3 Non-Functional Requirements8

3.4 Hardware and Software Requirements8

3.5 Use Case Model8

APPLICATION DESIGN10

4.1 Application Structure10

4.2 Data Model10

4.3 Form Design11

4.4 Process Flow Design15

4.5 Screen and Dashboard Design16

4.6 Access Design16

IMPLEMENTATION ON ZOHO CREATOR18

5.1 Creating the Application18

5.2 Forms18

5.3 Formula Fields and Field Rules19

5.4 Reports21
5.5 Pages and Dashboards24
5.6 Workflows and Automation25
5.7 Users, Roles and Permissions27
5.8 Data Import, Export and Integration28
5.9 AI-Assisted Features29
5.10 Mobile Application30
TESTING31
6.1 Testing Approach31
6.2 Test Cases31
6.3 Problems Faced and Fixes32
6.4 User Acceptance Testing33
PUBLISHING, SHARING AND SECURITY34
7.1 Publishing the Application34
7.2 Security and Data Protection35
7.3 Administration and Maintenance35
RESULTS AND DISCUSSION36
8.1 Application Output36
8.2 Achievement of Objectives37
8.3 Comparison with the Existing System38
8.4 Limitations of the No-Code Approach38
CONCLUSION AND FUTURE ENHANCEMENTS39
9.1 Conclusion39
9.2 Future Enhancements39
9.3 Relevance to My Discipline39
REFERENCES40
APPENDIX A – ZOHO CREATOR NO-CODE FEATURE CHECKLIST41
APPENDIX B – COURSE OUTCOME MAPPING43
APPENDIX C – APPLICATION CONFIGURATION RECORD44
APPENDIX D – USER MANUAL47
D.1 Accessing the Application47
D.2 Instructions for Receptionist47
D.3 Instructions for Doctor47
D.4 Instructions for Administrator47
D.5 Instructions for Patient47
APPENDIX E – EVALUATION RUBRIC49

APPENDIX F – VIVA-VOCE RECORD SHEET51

LIST OF TABLES

Table 2.1: Comparison of Development Approaches for the Proposed Application 5

Table 2.2: Summary of Literature Survey 5

Table 3.1: Users and Roles 7

Table 3.2: Functional Requirements 7

Table 3.3: Non-Functional Requirements 8

Table 3.4: Hardware and Software Requirements 8

Table 3.5: Use Case Description – Book Appointment 9

Table 3.6: Use Case Description – Approve Doctor Leave 9

Table 4.1: Forms and Their Relationships 11

Table 4.2: Field Specification – Department Form 12

Table 4.3: Field Specification – Doctor Form 12

Table 4.4: Field Specification – Patient Registration Form 12

Table 4.5: Field Specification – Appointment Form 13

Table 4.6: Field Specification – Consultation Record Form 14

Table 4.7: Field Specification – Billing Form 14

Table 4.8: Field Specification – Doctor Leave Request Form 15

Table 4.9: Field Specification – Patient Feedback Form 15

Table 4.10: Role–Permission Matrix 16

Table 5.1: Formula Fields and Rules Configured 20

Table 5.2: Reports Configured 21

Table 5.3: Workflows Configured 25

Table 6.1: Test Cases 31

Table 6.2: Issue Log 32

Table 6.3: User Acceptance Feedback 33

Table 7.1: Publishing Checklist 34

Table 7.2: Security Settings Applied 35

Table 8.1: Mapping of Objectives to Outcomes 37

Table 8.2: Existing System versus Proposed Application 38

Table A.1: No-Code Feature Coverage Checklist 41

Table B.1: Mapping of Course Outcomes to Report Chapters 43

Table C.1: Summary of Application Components 44

Table C.2: Workflow Settings in Detail – Patient Welcome Alert 45

Table C.3: Workflow Settings in Detail – Slot Clash Check 45

Table C.4: Workflow Settings in Detail – Appointment Confirmation 45

Table C.5: Workflow Settings in Detail – Consultation Completed 45
Table C.6: Workflow Settings in Detail – Doctor Leave Approval 45
Table C.7: Workflow Settings in Detail – Billing Discount Approval 46
Table C.8: Workflow Settings in Detail – Appointment Reminder 46
Table C.9: Workflow Settings in Detail – Daily No-Show Marker 46
Table E.1: Evaluation Rubric for the Individual Project 49
Table E.2: Performance Level Descriptors 49
Table F.1: Viva-Voce Questions and Responses 51
Table F.2: Marks Summary 51

LIST OF FIGURES

Figure 2.1: As-Is Process Flow of the Existing System 4
Figure 3.1: Use Case Diagram of the Application 9
Figure 4.1: Overall Structure of the Application 10
Figure 4.2: Data Model Showing Forms and Their Links 11
Figure 4.3: Process Flow for Patient Registration to Billing 16
Figure 4.4: Menu Structure and Dashboard Wireframe 16
Figure 5.1: Application Home Screen and Menu 18
Figure 5.2: Patient Registration Form in the Form Builder 19
Figure 5.3: Patient Registration Form as Seen by the User 19
Figure 5.4: Field Rule / Formula Configuration 21
Figure 5.5: All Appointments Report 22
Figure 5.6: Appointment Board (Kanban Report) 23
Figure 5.7: Appointment Calendar Report 23
Figure 5.8: Revenue Summary (Pivot Report) 24
Figure 5.9: Admin Dashboard Page 25
Figure 5.10: Appointment Confirmation in the Workflow Builder 26
Figure 5.11: Appointment Reminder Schedule Configuration 27
Figure 5.12: Doctor Leave Approval Configuration 27
Figure 5.13: Roles and Permissions Configuration 28
Figure 5.14: Data Import Setup 29
Figure 5.15: AI Application Builder in Use 30
Figure 5.16: Application on the Mobile App 30
Figure 7.1: Publish / Share Settings 34
Figure 8.1: Scenario Step 1 – Receptionist Books the Appointment 36
Figure 8.2: Scenario Step 2 – Administrator Approves a Doctor Leave Request 37
Figure 8.3: Scenario Step 3 – Doctor Records the Consultation 37

LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
CRUD	Create, Read, Update, Delete
CSV	Comma-Separated Values
FR	Functional Requirement
HMIS	Hospital Management Information System
KPI	Key Performance Indicator
NCDP	No-Code Development Platform
NFR	Non-Functional Requirement
OPD	Out-Patient Department
QR	Quick Response (code)
RBAC	Role-Based Access Control
SDLC	Software Development Life Cycle
SMS	Short Message Service
UAT	User Acceptance Testing
UHID	Unique Health Identification Number
UI	User Interface
Zia	Zoho's AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

A hospital is a service organisation in which thousands of small administrative transactions must be completed correctly before a single minute of clinical care can begin. The first of these transactions is the registration of the patient and the booking of a consultation slot with a doctor in the Out-Patient Department (OPD). In a typical multi-speciality hospital of about 150 beds, with 14 consultants spread over six departments, the front desk handles 200 to 300 registrations and appointments every day. Government and large private hospitals in India are increasingly adopting digital registration systems such as the e-Hospital platform of the National Informatics Centre [1] to reduce queues and paper handling.

Many small and medium hospitals and clinics, however, still depend on paper registration forms, bound registers, telephone calls and spreadsheets maintained by individual receptionists. Patient details are written more than once, appointment slots are tracked in a diary, doctors learn about their workload only when patients arrive, and the administration prepares monthly summaries by counting entries manually. Although a mechanical engineering student is not a hospital professional, the underlying problem — scheduling a limited resource (the doctor's time) against uncertain demand, and recording every job that passes through the service centre — is the same resource-scheduling and work-order problem that is met in maintenance workshops and production planning. This project therefore applies the thinking of industrial engineering, namely capacity planning, queue reduction and record keeping, to hospital OPD operations.

The project, titled **Hospital Patient Registration and Appointment System**, digitises this process on the Zoho Creator no-code platform. The application is designed for a hospital named "MediCare Hospital" (a representative organisation), with four groups of users: the hospital administrator, front-desk receptionists, consultant doctors and the patients themselves.

1.2 Problem Statement

The registration and appointment process of the hospital is paper-based and depends on the memory and diligence of individual staff members. It results in duplicate patient records, double booking of doctor slots, long waiting times, patients who forget their appointments, delays in approving doctors' leave and billing discounts, and no timely summary for the administration. A low-cost digital system that can be built and modified by non-programmers is required.

- **Duplicate and lost records:** the same patient is registered again at every visit on a fresh paper form, and files are misplaced or illegible.
- **Double booking and long queues:** slots are tracked in a diary, so two patients may be given the same doctor and time, while other slots stay unused.
- **No reminders:** patients are not reminded of appointments, causing a high number of no-shows and wasted consultation time.
- **Slow approvals:** doctor leave and billing discounts are approved on paper, so receptionists do not know which doctors are available.
- **No management summary:** the administrator has no real-time view of appointments, revenue and cancellations.

1.3 Objectives

- To digitise patient, doctor, department, appointment, consultation, billing, leave and feedback records using eight Zoho Creator forms with validations.
- To provide ten reports (list, Kanban, calendar, chart, pivot and spreadsheet) and two pages, including an administrator dashboard for daily monitoring.
- To automate confirmations, reminders, status updates and approvals using eight workflows of normal (form), schedule and approval types built in the visual workflow builder.
- To restrict access to data based on four user roles: Administrator, Receptionist, Doctor and Patient.
- To publish the application for web and mobile use by hospital staff and patients, and to evaluate it through test cases and user acceptance testing.

1.4 Scope of the Project

In scope: patient registration with a unique health identification number, doctor and department master data, appointment booking with slot validation, confirmation and reminder messages, consultation records with prescriptions, billing with discount approval, doctor leave approval, patient feedback, dashboards and reports, role-based access, data import from Excel, and use through the web browser, the Zoho Creator mobile app and a patient portal.

Out of scope: in-patient (ward) management, pharmacy stock, laboratory instrument integration, insurance claim processing, electronic health record exchange with external systems, and online payment gateway integration. The application records the payment mode and status only. The application is a student project that uses sample data; it is not intended to store real patient information.

1.5 Why No-Code?

A no-code platform allows a domain expert who is not a software engineer — a "citizen developer" — to build a working business application by dragging and dropping fields, setting properties and drawing workflows, without writing programs. Zoho Creator provides forms, reports, pages, workflow automation, approvals, role-based security, hosting and a mobile app in one environment [2]. Industry analysts regard such low-code and no-code application platforms as a major route to faster digital transformation, because business applications can be delivered in weeks instead of months [3], [4].

For this project the advantages are specific. No programming is needed, so a mechanical engineering student can build a complete hospital front-office system. Hosting, backup, security patches and the mobile app are provided by the platform. Changes such as a new field or a new department can be made in minutes. Built-in notification actions replace the manual telephone reminders. Finally, role-based access and audit history help to protect sensitive patient data. No Deluge scripting has been used anywhere in the application.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related literature on no-code platforms. Chapter 3 presents the requirement analysis. Chapter 4 describes the application design. Chapter 5 details how the application was built on Zoho Creator. Chapter 6 covers

testing. Chapter 7 discusses publishing, sharing and security. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing system a patient arrives at the registration counter and is given a paper form. The receptionist copies the details into a bound register and issues a hand-written file card. The patient then waits in a common queue because no time slot is allotted. The doctor consults the patient and writes the diagnosis and prescription on the file. After the consultation the patient goes to the cash counter where the bill is written manually, and any discount is approved verbally by the administrator. At month end the administrator counts the register entries and prepares summaries in a spreadsheet. Doctor leave is requested on paper, so the front desk is often not aware of a doctor's absence until the patient has arrived. Figure 2.1 shows this as-is flow.

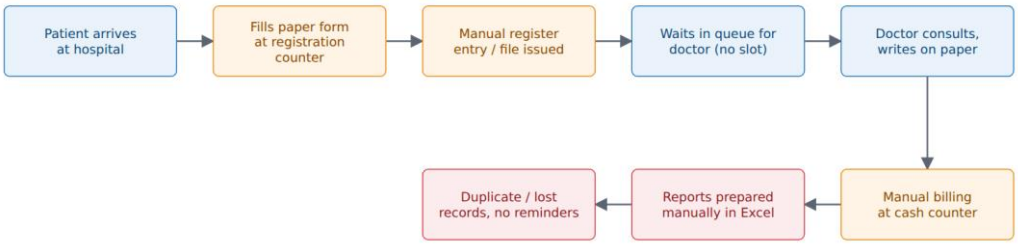


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Patient details are written repeatedly, leading to duplicate, incomplete or illegible records and no single patient identity number.
- No slot validation exists, so double booking and overcrowding of popular doctors are common.
- No automatic confirmation or reminder is sent, which increases missed appointments.
- Approvals for doctor leave and bill discounts are verbal or on paper and cannot be tracked.
- Preparing daily, weekly and monthly summaries is time consuming and error prone, and there is no real-time dashboard.

- Records cannot be accessed securely from a mobile device, and there is no role-based protection of patient data.

2.3 No-Code, Low-Code and Traditional Development

Low-code development platforms differ widely in the level of abstraction and the extent of coding they require [5]. Table 2.1 compares the three approaches for the proposed hospital application.

Table 2.1: Comparison of Development Approaches for the Proposed Application

Criterion	Traditional (Coding)	Low-Code	No-Code (This Project)
Programming skill needed	High – web, database and mobile programming	Medium – scripting for custom logic	None – drag-and-drop and settings
Development time	3 – 6 months	4 – 8 weeks	1 – 2 weeks
Hosting & mobile app	Separate servers and separate app to be built	Platform hosting; mobile app provided	Built-in hosting and ready mobile app
Ease of making changes	Needs a developer and redeployment	Moderate	Immediate through the builders
Flexibility / customisation	Unlimited	High	Moderate – limited to platform features
Cost	High (developers, servers)	Medium (licences + developer)	Low (platform subscription / education plan)

2.4 Literature Survey

Five sources were reviewed: platform documentation, industry analyst reports, academic papers on low-code platforms, and an existing hospital information system. Table 2.2 summarises them.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Zoho Corporation, 2026 [2]	Zoho Creator Help Documentation	Describes forms, reports, pages, workflows, approvals, schedules and roles that are configured visually.	Primary reference for every component built in Chapter 5.
2	Gartner, Inc., 2023 [3]	Magic Quadrant for Enterprise Low-Code Application Platforms	Evaluates low-code platforms and indicates rapid adoption for business applications.	Justifies the choice of a no-code platform.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
3	C. Richardson and J. R. Rymer, 2016 [4]	Vendor Landscape: The Fractured, Fertile Terrain of Low-Code Application Platforms	Introduces the low-code category and notes faster delivery with visual tools.	Supports faster delivery of the OPD application.
4	A. Sahay et al., 2020 [5]	Supporting the Understanding and Comparison of Low-Code Development Platforms	Compares features of low-code platforms to help users select one.	Guides the comparison in Table 2.1.
5	R. Waszkowski, 2019 [6]	Low-code platform for automating business processes in manufacturing	Reports the automation of a manufacturing business process with a low-code platform.	Shows the engineering use of such platforms for workflow automation.
6	D. Di Ruscio et al., 2022 [7]	Low-code development and model-driven engineering: Two sides of the same coin?	Discusses how low-code relates to model-based design of applications.	Supports the form and data-model driven design in Chapter 4.
7	National Informatics Centre, 2026 [1]	e-Hospital (Hospital Management Information System)	A government HMIS that offers online registration and appointment services.	Existing system benchmark for the proposed application.
8	Zoho Corporation, 2026 [8]	Zoho Creator Learning Centre	Provides tutorials and sample applications for building apps without code.	Used for learning workflows, approvals and role settings.

2.5 Proposed System

The proposed system is a Zoho Creator application with eight forms, ten reports, two pages and eight workflows. A patient is registered once and receives a unique health identification number (UHID). Appointments are booked by selecting the patient, department, doctor, date and slot; a workflow checks the slot and rejects double booking. Confirmations are sent immediately and reminders one day before the appointment. After the consultation the appointment is closed automatically and a bill is generated. Leave requests and large discounts pass through approval processes, and a daily schedule marks unattended appointments as no-shows. The administrator sees the live dashboard. Each limitation of Section 2.2 is therefore addressed: duplicates by the UHID and unique phone check, double booking by slot validation, missed visits by reminders, untracked approvals by approval processes, and the absence of summaries by dashboards and reports.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Users and Roles

Table 3.1: Users and Roles

Role	Who (in the organisation)	What the Role Does in the Application
Administrator	Hospital administrator / OPD manager	Manages users, departments, doctors and master data; approves leave and discounts; views dashboard and all reports.
Receptionist	Front-desk staff	Registers patients, books and reschedules appointments, prepares bills, views calendar and Kanban board.
Doctor	Consultant doctors	Views own appointments, records consultations and prescriptions, applies for leave.
Patient	Registered patients (portal users)	Books appointments, views own appointments and bills, submits feedback.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The application shall allow registration of a patient with mandatory fields and a unique phone number.	Receptionist, Administrator	High
FR-02	The application shall generate the UHID, appointment ID and bill number automatically and calculate the patient's age from the date of birth.	All	High
FR-03	The application shall allow booking of an appointment by patient, department, doctor, date and time slot, and shall block a slot that is already booked.	Receptionist, Patient	High
FR-04	The application shall send an email and a notification to the patient when an appointment is booked.	System	Medium
FR-05	The application shall remind the patient one day before a confirmed appointment.	System	Medium
FR-06	The application shall allow the doctor to record diagnosis and a multi-row prescription for an appointment.	Doctor	High
FR-07	The application shall mark the appointment as completed and generate a bill when a consultation record is saved.	System	High
FR-08	The application shall calculate the net bill amount and send a discount above 10% for approval by the Administrator.	Receptionist, Administrator	High
FR-09	The application shall send a doctor's leave request for approval to the Administrator.	Doctor, Administrator	Medium
FR-10	The application shall mark unattended appointments as No-Show at the end of each day.	System	Medium

FR ID	Requirement	Role(s)	Priority
FR-11	The application shall show a dashboard of KPIs and charts to the Administrator.	Administrator	Medium
FR-12	The application shall provide list, Kanban, calendar, chart, pivot and spreadsheet reports.	Administrator, Receptionist	Medium
FR-13	The application shall restrict data and menus according to the role of the user.	All	High
FR-14	The application shall allow patients to view their own appointments and submit feedback through the portal.	Patient	Medium
FR-15	The application shall allow import of doctor master data from Excel and export of reports to PDF / Excel.	Administrator	Low

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Usability	A new receptionist shall be able to register a patient and book an appointment without training, in under three minutes.
NFR-02	Security	Users shall see only the data allowed for their role; patients shall see only their own records.
NFR-03	Accessibility	The application shall work on desktop browsers and the Zoho Creator mobile app.
NFR-04	Reliability	The application shall be hosted on the platform with automatic backup, and workflows shall run without manual intervention.
NFR-05	Performance	Forms and reports shall load within three seconds for up to 5,000 records.
NFR-06	Maintainability	An administrator shall be able to add a field, department or doctor without any programming.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator – Education / Free trial Edition
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
Mobile	Zoho Creator mobile app (Android / iOS)
Other Zoho services	Zoho Mail (for email notifications); built-in export to Excel and PDF
Client hardware	Any internet-enabled desktop, laptop or smartphone

3.5 Use Case Model

The use case diagram in Figure 3.1 shows the four actors and the main functions of the application. Tables 3.5 and 3.6 describe two key use cases in detail.

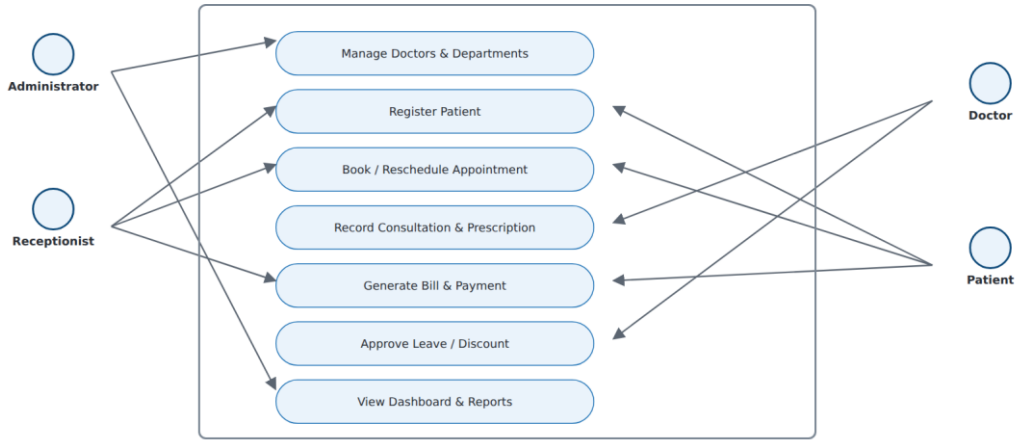


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Book Appointment

Use Case ID	UC-01
Use Case Name	Book Appointment
Actor(s)	Receptionist, Patient
Pre-condition	The patient is registered and the doctor is active and not on approved leave.
Main Flow	1. The actor opens the Appointment form. 2. Selects the patient, department and doctor. 3. Selects the date and time slot. 4. Submits the form. 5. The slot is validated, the appointment ID is generated and a confirmation is sent.
Alternate Flow	If the slot is already booked, an alert is shown and the record is not saved; the actor selects another slot.
Post-condition	An appointment with status Booked is saved and the patient is notified.
Related FR	FR-02, FR-03, FR-04

Table 3.6: Use Case Description – Approve Doctor Leave

Use Case ID	UC-02
Use Case Name	Approve Doctor Leave
Actor(s)	Doctor, Administrator
Pre-condition	The doctor is logged in and has an active profile.
Main Flow	1. The doctor submits a Doctor Leave Request. 2. The approval process routes it to the Administrator. 3. The Administrator approves or rejects with a comment. 4. The status is updated and the doctor is informed by email.
Alternate Flow	If there is no response within 24 hours, a reminder is sent to the Administrator.
Post-condition	The leave request carries the final status Approved or Rejected.
Related FR	FR-09

CHAPTER 4
APPLICATION DESIGN

4.1 Application Structure

Figure 4.1 shows the overall structure. Four groups of users access the application through the web, the Zoho Creator mobile app and the patient portal. The application contains eight forms, ten reports, two pages and eight workflows, and sends email, SMS and push notifications. The menu is organised into sections for Patients, Doctors, Appointments, Consultations, Billing, Administration and Reports.

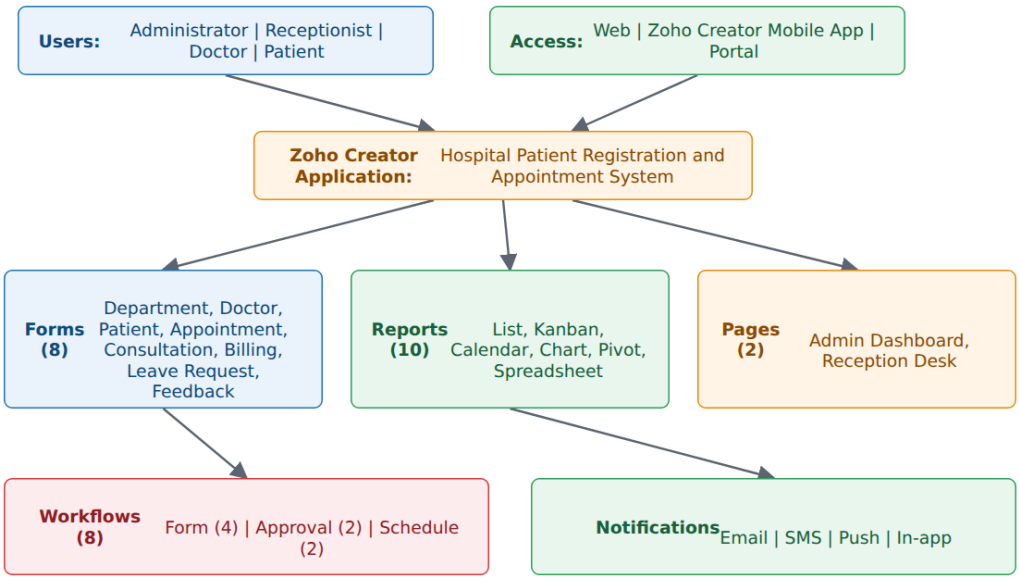


Figure 4.1: Overall Structure of the Application

4.2 Data Model

Eight forms are linked through lookup fields. A Department has many Doctors; a Doctor and a Patient each have many Appointments; an Appointment has one Consultation Record, one Bill and optionally one Feedback; and a Doctor has many Leave Requests. The prescription rows are stored in a subform inside the Consultation Record. Figure 4.2 shows the data model and Table 4.1 lists the relationships.

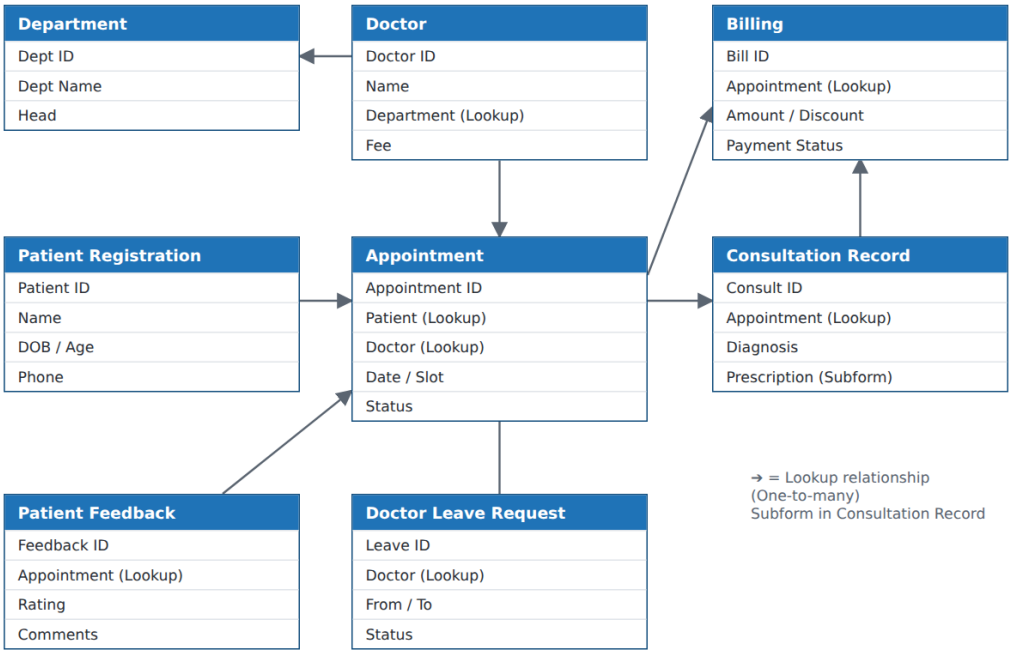


Figure 4.2: Data Model Showing Forms and Their Links

Table 4.1: Forms and Their Relationships

Form	Purpose	Linked To	Link Type	Configured Using
Department	Master list of departments	Doctor, Appointment	One department – many doctors	Lookup field
Doctor	Master list of doctors	Appointment, Leave Request	One doctor – many appointments	Lookup field
Patient Registration	Patient master record	Appointment	One patient – many appointments	Lookup field
Appointment	Each OPD booking	Consultation, Billing, Feedback	One appointment – one consultation / bill	Lookup field
Consultation Record	Diagnosis and prescription	Prescription rows	One record – many rows	Subform
Billing	Charges and payment	Appointment	Many bills – one appointment	Lookup field
Doctor Leave Request	Leave applications	Doctor	Many requests – one doctor	Lookup field
Patient Feedback	Post-visit feedback	Appointment	Many – one	Lookup field

4.3 Form Design

Each form is specified in the tables below using the field-type names of Zoho Creator. In total 24 distinct field types are used across the eight forms: Auto Number, Single Line, Multi Line, Name, Phone, Email, Date, Date-Time, Time, Drop Down, Radio,

Multi Select, Decision Box, Lookup, Subform, Number, Decimal, Currency, Percent, Formula, Image, File Upload, Signature and Address.

Table 4.2: Field Specification – Department Form

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Department ID	Auto Number	—	Prefix DEPT-	System generated
Department Name	Single Line	Yes	Unique	No duplicate departments
Head of Department	Single Line	Yes	—	Senior consultant
Location	Drop Down	Yes	Block A / B / C	Building block
Extension Number	Number	No	Min 3 digits	Internal phone
Description	Multi Line	No	—	Services offered
Status	Radio	Yes	Default: Active	Active / Inactive

Table 4.3: Field Specification – Doctor Form

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Doctor ID	Auto Number	—	Prefix DOC-	System generated
Doctor Name	Name	Yes	—	First and last name
Department	Lookup	Yes	Looks up Department	Filters doctors in Appointment
Qualification	Single Line	Yes	—	e.g., MBBS, MD
Phone	Phone	Yes	Unique	Contact number
Email	Email	Yes	Unique	Used for notifications
Consultation Fee	Currency	Yes	Min 0	Auto-filled in Appointment
Available Days	Multi Select	Yes	Mon – Sat	OPD days
Slot Start / End Time	Time	Yes	Default 09:00 / 17:00	OPD hours
Photo	Image	No	—	Profile picture
Status	Drop Down	Yes	Default: Active	Active / On Leave / Inactive

Table 4.4: Field Specification – Patient Registration Form

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Patient ID	Auto Number	—	Prefix UHID-	System generated UHID

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Patient Name	Name	Yes	—	First and last name
Date of Birth	Date	Yes	Validation: ≤ today	Used for age
Age	Formula	—	Years from Date of Birth	Auto-calculated
Gender	Radio	Yes	Male / Female / Other	—
Blood Group	Drop Down	Yes	A+, A−, B+, B−, O+, O−, AB+, AB−	—
Phone	Phone	Yes	Unique	Duplicate check
Email	Email	No	—	For confirmations
Address	Address	Yes	—	Residential address
Photo	Image	No	Camera capture	Mobile use
Known Allergies	Multi Line	No	—	Safety information
Consent for Treatment	Decision Box	Yes	—	Must be ticked
Registered On	Date-Time	—	Default: current time	Audit

Table 4.5: Field Specification – Appointment Form

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Appointment ID	Auto Number	—	Prefix APT-	System generated
Patient	Lookup	Yes	Looks up Patient Registration	One patient – many appointments
Department	Lookup	Yes	Looks up Department	—
Doctor	Lookup	Yes	Filtered by Department	Active doctors only
Appointment Date	Date	Yes	Validation: ≥ today	No past bookings
Time Slot	Drop Down	Yes	09:00 – 16:30, 30 min	Slot clash checked by workflow
Consultation Fee	Currency	No	Set from Doctor	Read-only
Visit Type	Radio	Yes	Default: New	New / Follow-up
Reason for Visit	Multi Line	No	—	—
Status	Drop Down	Yes	Default: Booked	Booked / Confirmed / Completed / Cancelled / No-Show
Cancellation Reason	Multi Line	Conditional	Shown if Status = Cancelled	Field rule

Table 4.6: Field Specification – Consultation Record Form

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Consultation ID	Auto Number	—	Prefix CON-	System generated
Appointment	Lookup	Yes	Looks up Appointment	Links to patient and doctor
Consultation Date	Date	Yes	Default: today	—
Symptoms	Multi Line	Yes	—	—
Diagnosis	Multi Line	Yes	—	—
Temperature (°F)	Decimal	No	95 – 106	Vital sign
Prescription	Subform	No	Medicine, Dosage, Days, Instructions	Many rows per record
Lab Report	File Upload	No	PDF / image	—
Follow-up Required	Decision Box	No	—	Drives Follow-up Date
Follow-up Date	Date	Conditional	Shown if follow-up ticked	Field rule
Doctor Signature	Signature	No	—	Digital signature

Table 4.7: Field Specification – Billing Form

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Bill No	Auto Number	—	Prefix BILL-	System generated
Appointment	Lookup	Yes	Looks up Appointment	—
Consultation Fee	Currency	Yes	Min 0	From Appointment
Other Charges	Currency	No	Default 0	Lab / medicine
Discount %	Percent	No	Default 0	> 10 goes for approval
Discount Amount	Formula	—	(Fee + Charges) × Discount % / 100	Auto-calculated
Net Amount	Formula	—	Fee + Charges − Discount Amount	Auto-calculated
Payment Mode	Radio	Yes	Cash / UPI / Card / Insurance	—
Payment Status	Drop Down	Yes	Default: Pending	Pending / Paid / Waived
Approval Status	Drop Down	No	Default: Not Required	Set by approval process
Bill Date	Date	Yes	Default: today	—

Table 4.8: Field Specification – Doctor Leave Request Form

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Leave ID	Auto Number	—	Prefix LV-	System generated
Doctor	Lookup	Yes	Looks up Doctor	—
Leave Type	Drop Down	Yes	Casual / Medical / Conference / Other	—
From Date	Date	Yes	Validation: ≥ today	—
To Date	Date	Yes	Validation: ≥ From Date	—
Total Days	Formula	—	(To Date − From Date) + 1	Auto-calculated
Reason for Leave	Multi Line	Conditional	Shown if Leave Type = Other	Field rule
Supporting Document	File Upload	No	PDF / image	—
Approval Status	Drop Down	No	Default: Pending	Set by approval process
Approver Comments	Multi Line	No	—	Filled by approver

Table 4.9: Field Specification – Patient Feedback Form

Field Label	Field Type	Mandatory	Properties (Validation / Default / Unique)	Remarks
Feedback ID	Auto Number	—	Prefix FB-	System generated
Appointment	Lookup	Yes	Looks up Appointment	—
Overall Rating	Radio	Yes	1 to 5	—
Waiting Time	Drop Down	Yes	< 15 min / 15–30 / > 30 min	—
Staff Behaviour	Drop Down	No	Poor / Average / Good / Excellent	—
Comments	Multi Line	No	—	—
Would Recommend	Decision Box	No	—	—
Submitted On	Date-Time	—	Default: current time	Audit

4.4 Process Flow Design

The end-to-end flow of a patient visit is shown in Figure 4.3. The steps that are automated by workflows are the slot validation, the confirmation message, the one-day reminder, the status update and bill generation after consultation, the discount approval, and the feedback request.

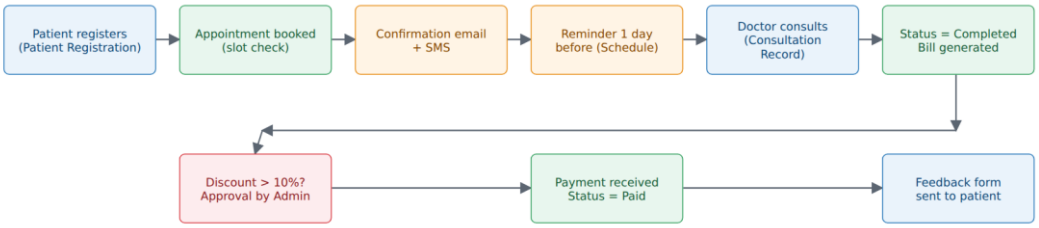


Figure 4.3: Process Flow for Patient Registration to Billing

4.5 Screen and Dashboard Design

Before building, the main dashboard was sketched as the wireframe of Figure 4.4. A left menu lists the forms and reports. The top band carries four KPI widgets (total patients, appointments today, pending bills and cancellations). Below them are a bar chart of appointments by department, a pie chart of appointment status and an embedded report of today's appointments.

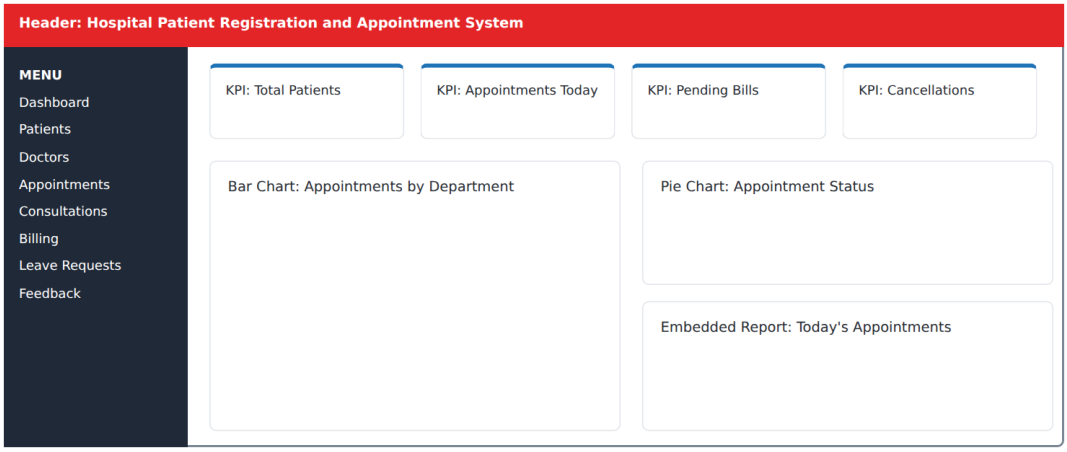


Figure 4.4: Menu Structure and Dashboard Wireframe

4.6 Access Design

Access is designed on the principle of least privilege. A = Add, E = Edit, V = View, D = Delete, — = No access.

Table 4.10: Role–Permission Matrix

Component	Administrator	Receptionist	Doctor	Patient
Form: Department	A E V D	V	V	—
Form: Doctor	A E V D	V	V (own)	—
Form: Patient Registration	A E V D	A E V	V	V (own)
Form: Appointment	A E V D	A E V	E V (own)	A V (own)
Form: Consultation Record	V	—	A E V (own)	—

Component	Administrator	Receptionist	Doctor	Patient
Form: Billing	A E V D	A E V	—	V (own)
Form: Doctor Leave Request	E V D	—	A V (own)	—
Form: Patient Feedback	V	V	—	A V (own)
Report: All Appointments	V	V	V (own)	—
Report: My Appointments	—	—	—	V (own)
Report: Revenue Summary	V	—	—	—
Page: Admin Dashboard	V	—	—	—
Page: Reception Desk	V	V	—	—

CHAPTER 5
IMPLEMENTATION ON ZOHO CREATOR

5.1 Creating the Application

The application was created from scratch using **Create Application → Start from scratch** and named "Hospital Patient Registration and Appointment System". The application sections and menu were arranged as Patients, Doctors, Appointments, Consultations, Billing, Administration and Reports. A red-and-white theme with a hospital icon was selected under **Settings → Theme**. The AI builder was used only to obtain an initial list of forms (Section 5.9); everything else was built manually. Figure 5.1 shows the home screen and menu.

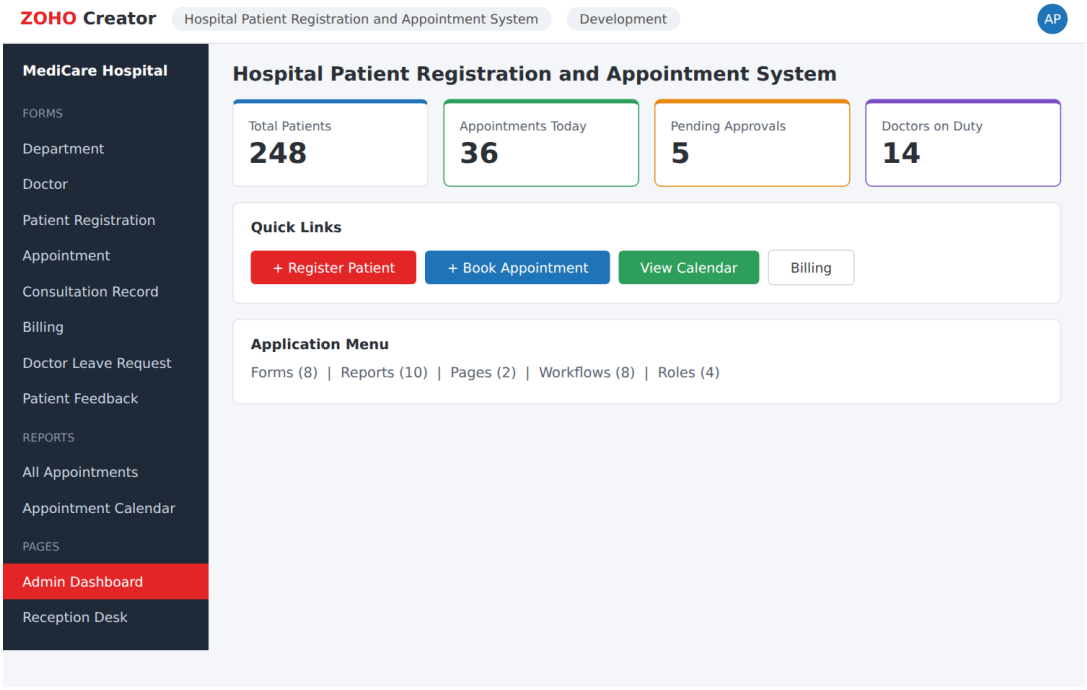


Figure 5.1: Application Home Screen and Menu

5.2 Forms

Eight forms were created in the form builder by dragging field types from the palette. Mandatory, unique and default-value properties were set for each field as given in Tables 4.2 to 4.9. Lookup fields connect Doctor to Department and Appointment to Patient, Doctor and Department. The Consultation Record form contains a Prescription subform in which the doctor can add any number of medicine rows. The Doctor lookup in the Appointment form is filtered by the selected department, so that only the doctors of that department appear. Success messages such as "Patient registered successfully"

and custom submit-button labels were also configured. Figures 5.2 and 5.3 show the Patient Registration form in the builder and as seen by the user.

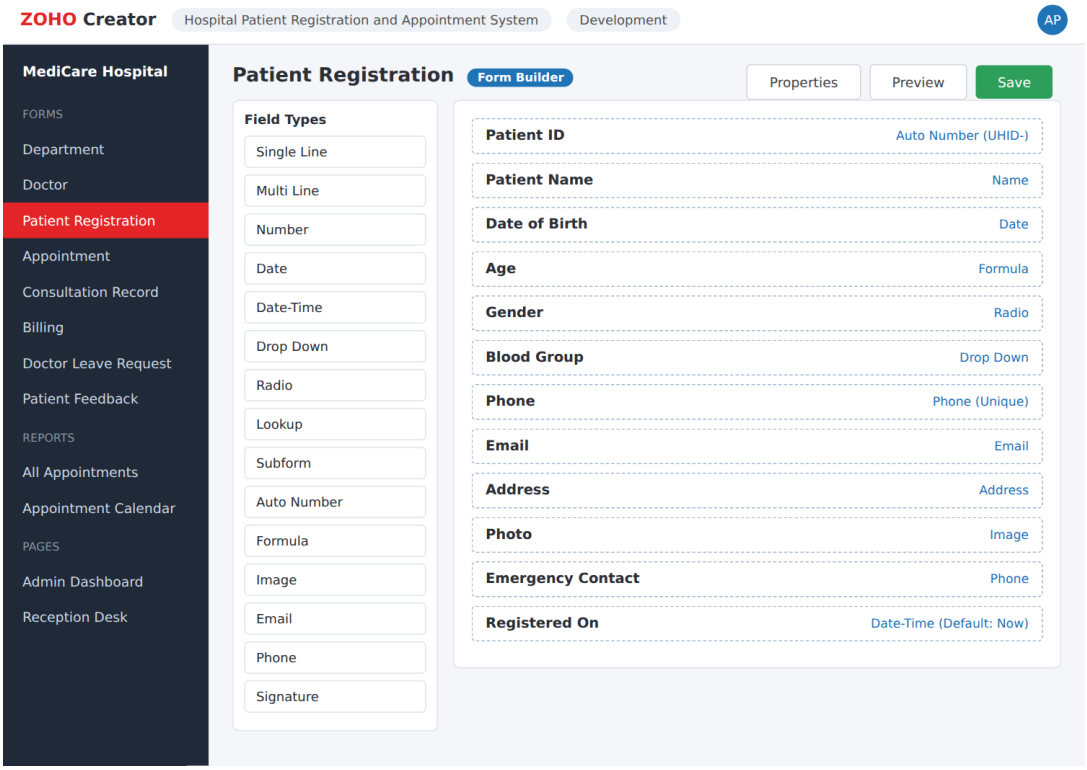


Figure 5.2: Patient Registration Form in the Form Builder

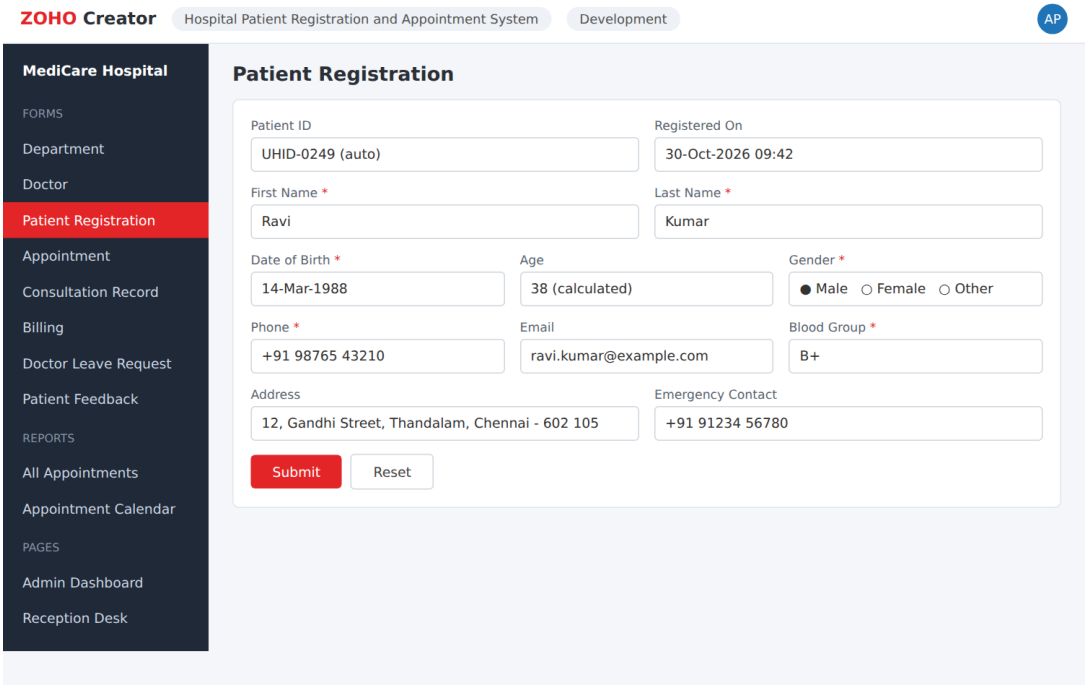


Figure 5.3: Patient Registration Form as Seen by the User

5.3 Formula Fields and Field Rules

No-code logic was added through formula fields, field rules and validation rules (Table 5.1). The Age field is calculated from the date of birth, the Total Days field from the leave dates, and the discount and net amounts from the charges. Field rules show or hide the Cancellation Reason, Follow-up Date and Reason for Leave fields according to the user's choice. Validation rules prevent a past appointment date and a leave end date earlier than the start date. Figure 5.4 shows the configuration.

Table 5.1: Formula Fields and Rules Configured

Name	Type	Form	Condition / Expression	Effect
Age	Formula field	Patient Registration	Years between Date of Birth and today	Auto-calculates age
Total Days	Formula field	Doctor Leave Request	(To Date − From Date) + 1	Auto-calculates leave days
Discount Amount	Formula field	Billing	(Fee + Other Charges) × Discount % / 100	Auto-calculates discount
Net Amount	Formula field	Billing	Fee + Other Charges − Discount Amount	Auto-calculates payable amount
Show Cancellation Reason	Field rule	Appointment	If Status = Cancelled	Show and require the reason
Show Follow-up Date	Field rule	Consultation Record	If Follow-up Required is ticked	Show Follow-up Date
Show Leave Reason	Field rule	Doctor Leave Request	If Leave Type = Other	Show Reason for Leave
Fill Consultation Fee	Field rule	Appointment	On selecting Doctor	Sets fee from Doctor record
Doctor by Department	Lookup filter	Appointment	Doctor.Department = selected Department	Shows only matching doctors
Appointment Date Check	Validation rule	Appointment	Appointment Date ≥ Today	Error: "Past dates are not allowed"
Birth Date Check	Validation rule	Patient Registration	Date of Birth ≤ Today	Error: "Date of birth cannot be in the future"
Leave Date Check	Validation rule	Doctor Leave Request	To Date ≥ From Date	Error: "To Date cannot be earlier than From Date"

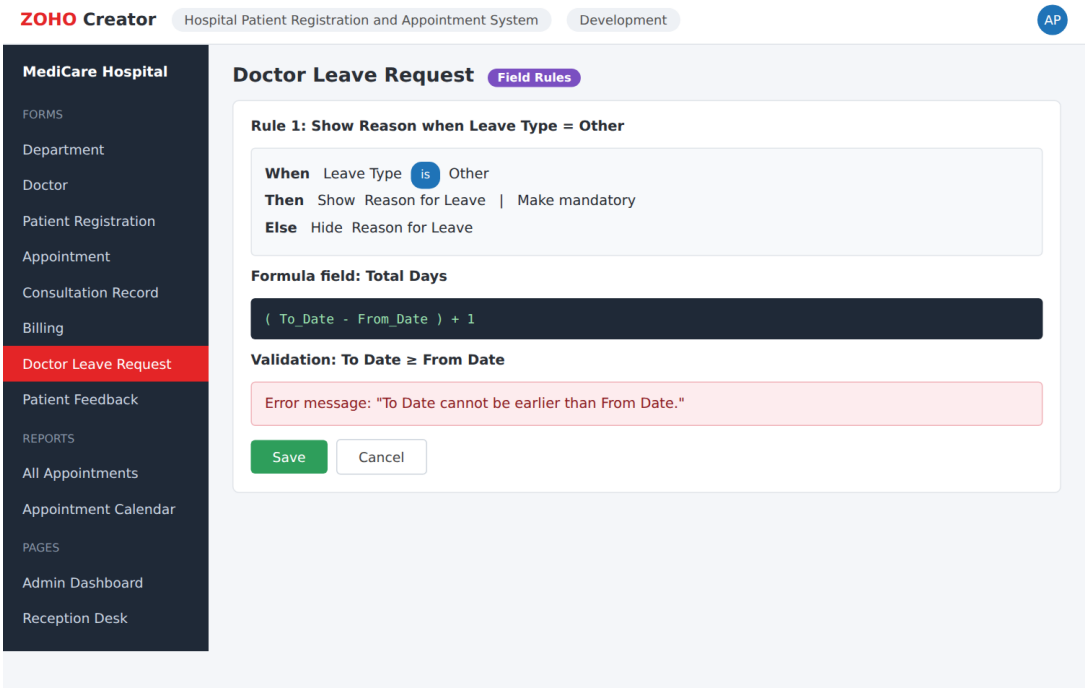


Figure 5.4: Field Rule / Formula Configuration

5.4 Reports

Ten reports of six types were configured (Table 5.2). Filters, grouping, sorting and conditional formatting were applied through the report settings. The All Appointments list report colours rows by status; the Appointment Board is a Kanban report with columns for each status so that the receptionist can drag a card from Booked to Confirmed; the Appointment Calendar shows the day-wise load; and the Revenue Summary pivot report totals the net amount by payment mode and month. Figures 5.5 to 5.8 show four of them.

Table 5.2: Reports Configured

Report Name	Type	Based on Form	Filter / Group / Sort	Formatting / Buttons	Roles
All Appointments	List	Appointment	Group by Date; sort by Time; filter by Status	Colour by status (Cancelled red, Completed green)	Administrator, Receptionist, Doctor (own)
Patient Directory	List	Patient Registration	Sort by Name; filter by Blood Group	Quick-view record template	Administrator, Receptionist
Doctor Directory	List	Doctor	Group by Department	Photo column	Administrator, Receptionist
Appointment Board	Kanban	Appointment	Columns by Status	Card shows Patient and Doctor	Administrator, Receptionist

Report Name	Type	Based on Form	Filter / Group / Sort	Formatting / Buttons	Roles
Appointment Calendar	Calendar	Appointment	Date = Appointment Date	Colour by Status	Administrator, Receptionist, Doctor
Appointments by Department	Chart (Bar)	Appointment	X: Department; Y: Count	—	Administrator
Revenue Summary	Pivot	Billing	Rows: Payment Mode; Columns: Month; Sum of Net Amount	Totals row	Administrator
Billing Sheet	Spreadsheet	Billing	Sort by Bill Date	Inline editing	Administrator, Receptionist
Pending Leave Requests	List	Doctor Leave Request	Filter: Approval Status = Pending	Approve / Reject buttons	Administrator
My Appointments	List	Appointment	Filter: Patient = logged-in user	Cancel button	Patient

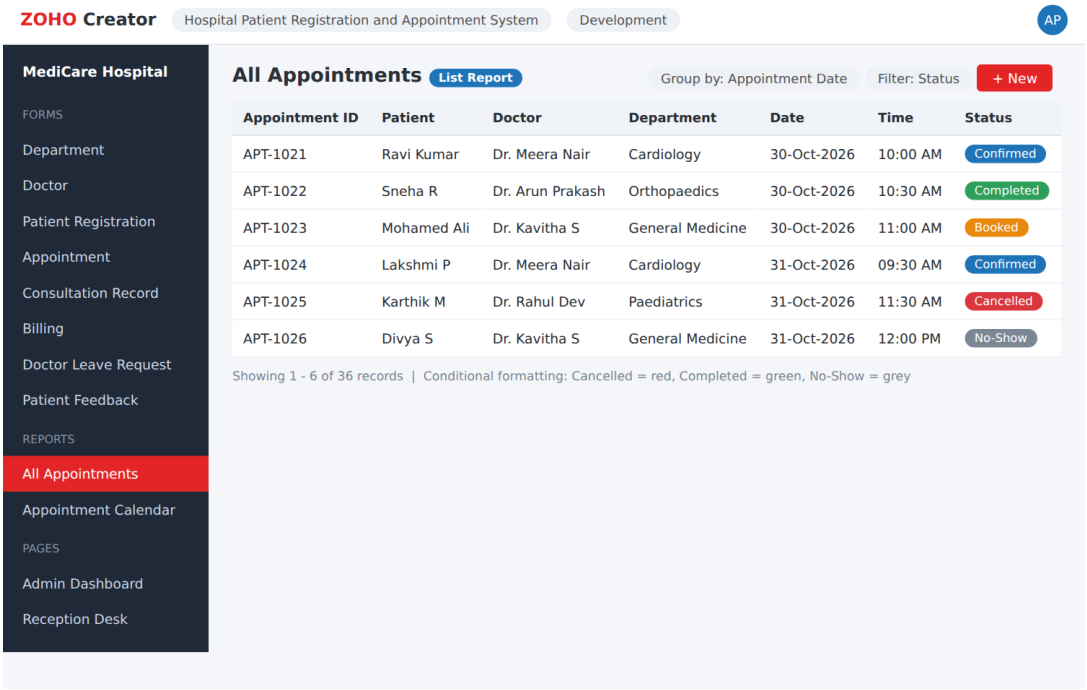


Figure 5.5: All Appointments Report

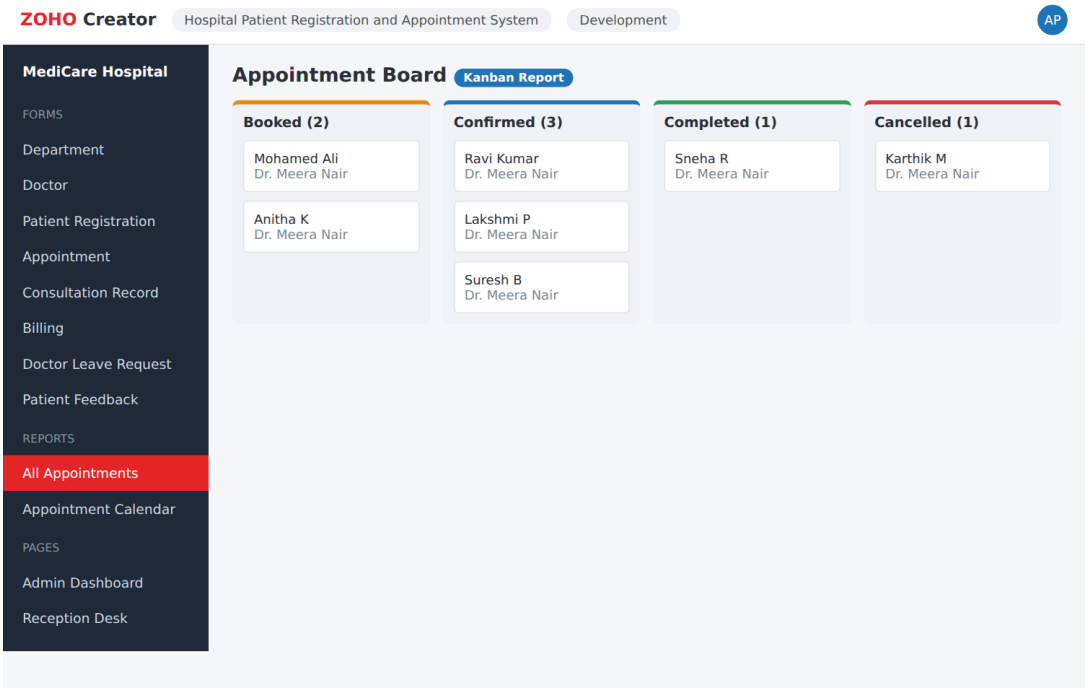


Figure 5.6: Appointment Board (Kanban Report)

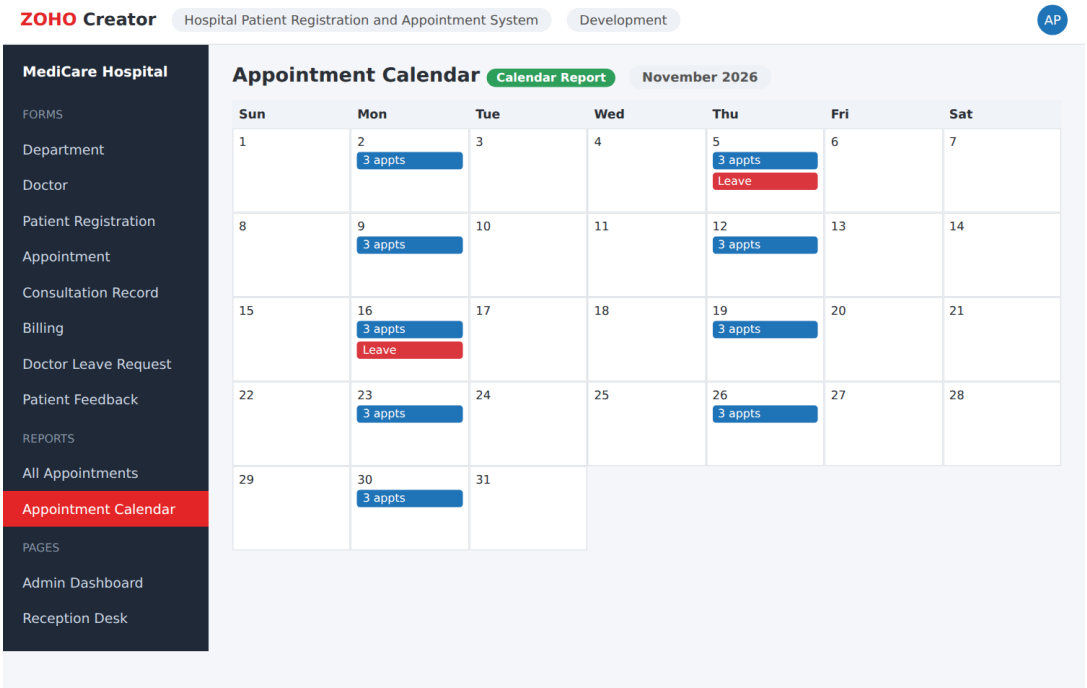


Figure 5.7: Appointment Calendar Report

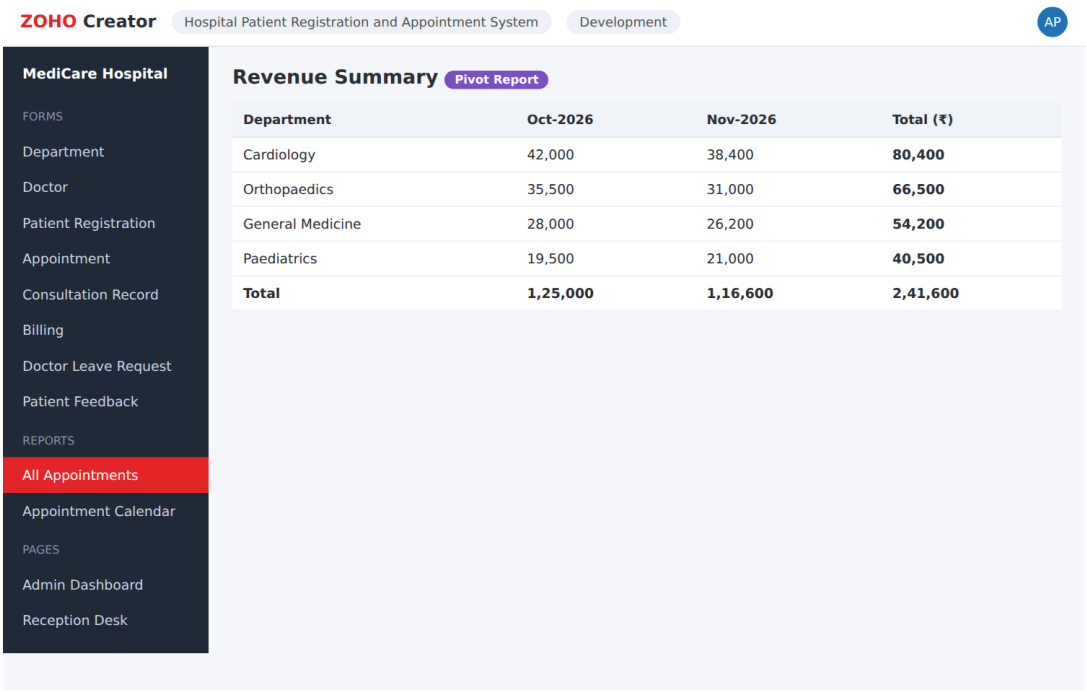


Figure 5.8: Revenue Summary (Pivot Report)

5.5 Pages and Dashboards

Two pages were built with the page builder. The **Admin Dashboard** is visible only to the Administrator. It contains four KPI widgets (Total Patients, Appointments Today, Pending Bills and Cancelled This Week), a bar chart of appointments by department and a doughnut chart of appointment status. The **Reception Desk** page is used by receptionists; it embeds the Patient Registration and Appointment forms, buttons for the Patient Directory and Billing Sheet, and a list of today's appointments, so that all front-desk tasks can be completed from one screen. Figure 5.9 shows the dashboard.



Figure 5.9: Admin Dashboard Page

5.6 Workflows and Automation

Eight workflows were configured in the visual workflow builder: four form workflows (normal), two approval processes and two schedules (Table 5.3). No Deluge script was written. Detailed settings of each workflow are recorded in Appendix C. Figure 5.10 shows the Appointment Confirmation form workflow, Figure 5.11 the Appointment Reminder schedule and Figure 5.12 the Doctor Leave Approval process.

Table 5.3: Workflows Configured

Workflow Name	Type	Trigger	Condition	Action(s)
Patient Welcome Alert	Form workflow (Normal)	Patient Registration – On Add (success)	Email is not empty	Send welcome email with UHID to patient; send in-app notification to Receptionist
Slot Clash Check	Form workflow (Normal)	Appointment – On Validate (Add / Edit)	Another record has same Doctor, Date, Time Slot and Status ≠ Cancelled	Show alert "Slot already booked" and stop submission
Appointment Confirmation	Form workflow (Normal)	Appointment – On Add (success)	Status = Booked	Email to patient and doctor; SMS / push notification; Status set to Confirmed
Consultation Completed	Form workflow (Normal)	Consultation Record – On Add (success)	Appointment is not empty	Update Appointment Status = Completed; create Billing record with fee; request feedback

Workflow Name	Type	Trigger	Condition	Action(s)
Doctor Leave Approval	Approval	Doctor Leave Request – On Add	Total Days ≥ 1	Administrator approves or rejects; status updated; email to doctor
Billing Discount Approval	Approval	Billing – On Add / Edit	Discount % > 10	Administrator approves or rejects; on rejection Discount % reset to 0
Appointment Reminder	Schedule (date-based)	1 day before Appointment Date, 9:00 AM	Status = Confirmed	Send email and push notification to patient
Daily No-Show Marker	Schedule (recurring)	Every day at 8:00 PM	Date = today and Status is Booked or Confirmed	Update Status = No-Show; email summary to Administrator

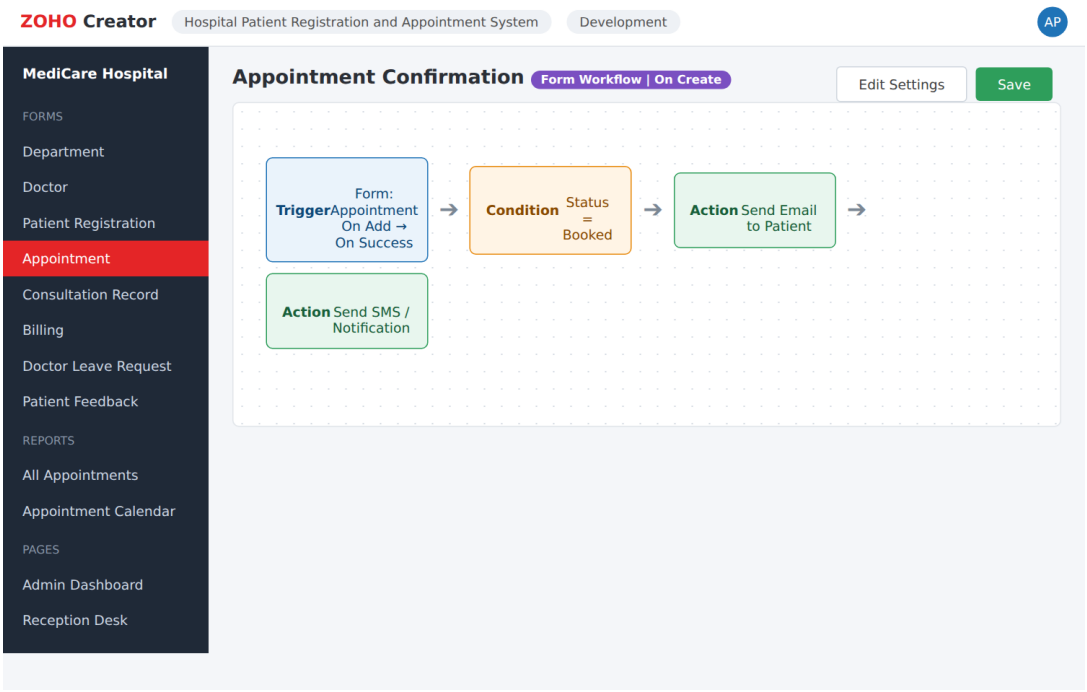


Figure 5.10: Appointment Confirmation in the Workflow Builder

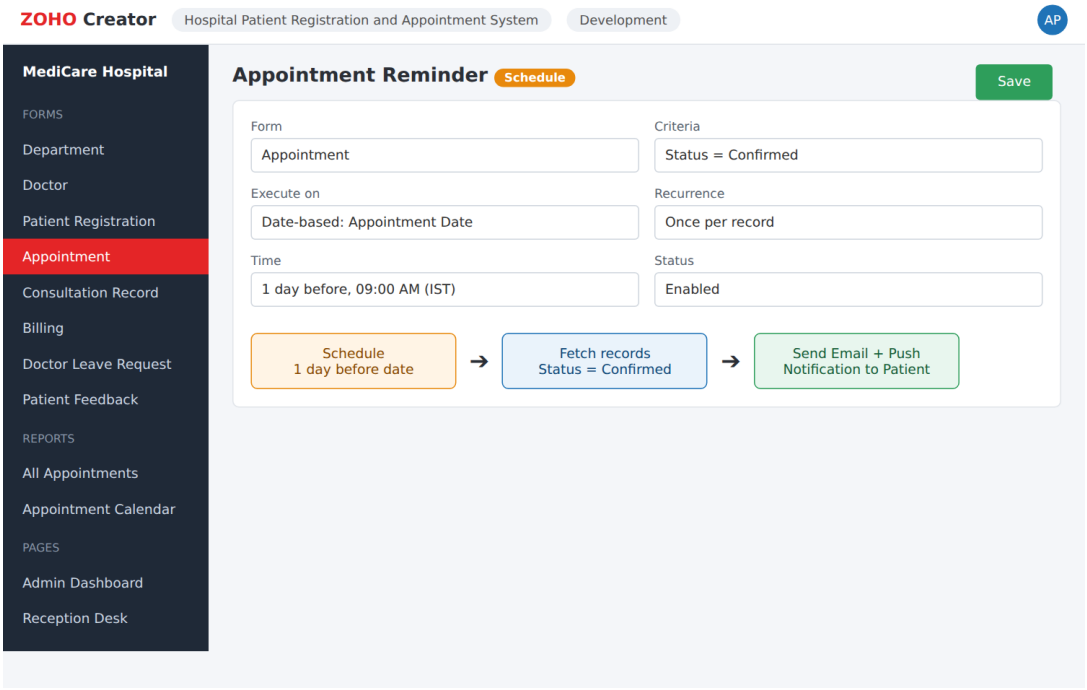


Figure 5.11: Appointment Reminder Schedule Configuration

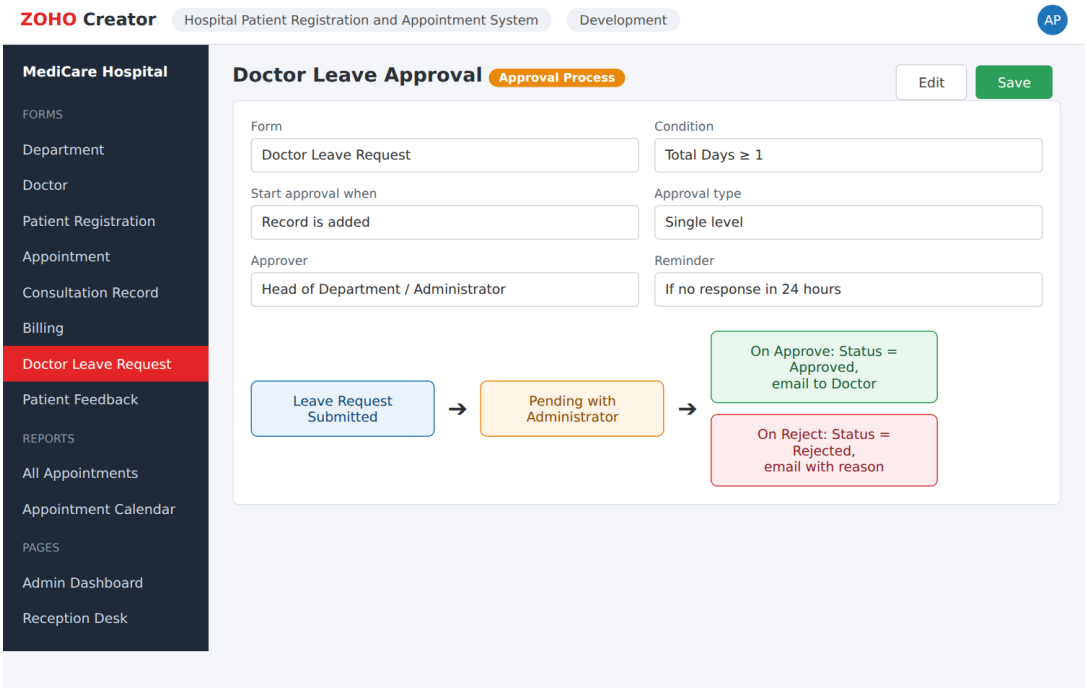


Figure 5.12: Doctor Leave Approval Configuration

5.7 Users, Roles and Permissions

Four roles — Administrator, Receptionist, Doctor and Patient — were created in **Users & Control** and the permissions of Table 4.10 were applied as permission sets. Record-level rules limit a doctor to appointments where the Doctor field matches the logged-in user, and a patient to records where the patient's email matches the login email. The

patient role is used through the customer portal. Figure 5.13 shows the roles and the permissions of the Receptionist role.

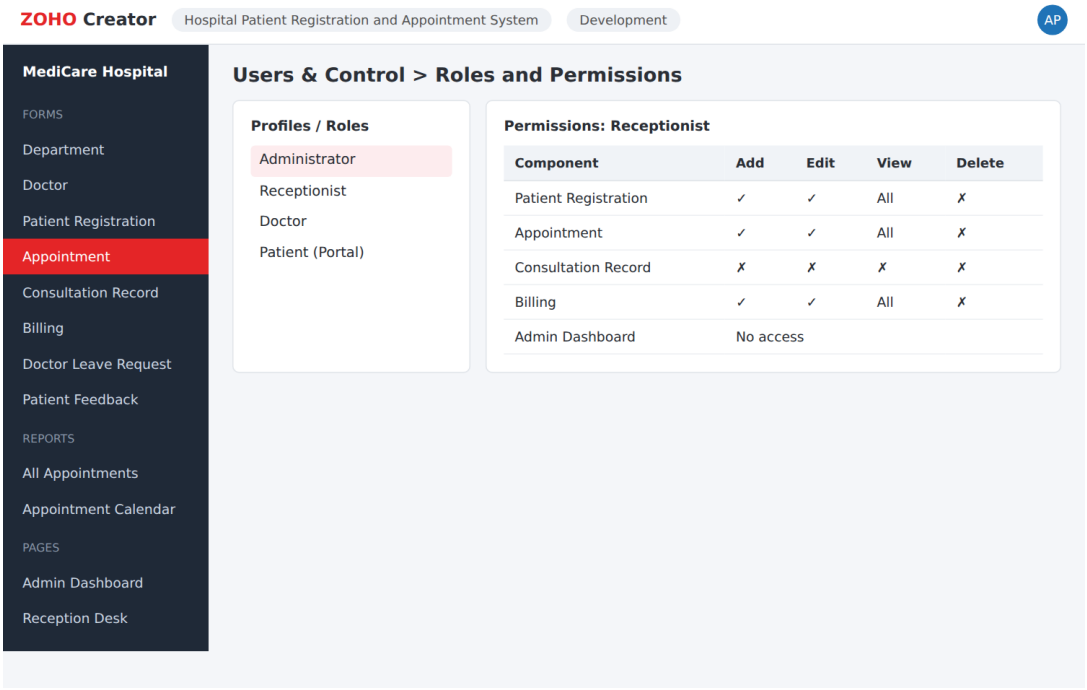


Figure 5.13: Roles and Permissions Configuration

5.8 Data Import, Export and Integration

The master list of 14 doctors was prepared in Excel and imported into the Doctor form using the **Import Data** option with column mapping (Figure 5.14). Reports can be exported to Excel, CSV and PDF from the report toolbar, and the link of the Doctor Directory report was shared in view-only mode. Email notifications are sent through the built-in integration with Zoho Mail, and no external connector or code was required.

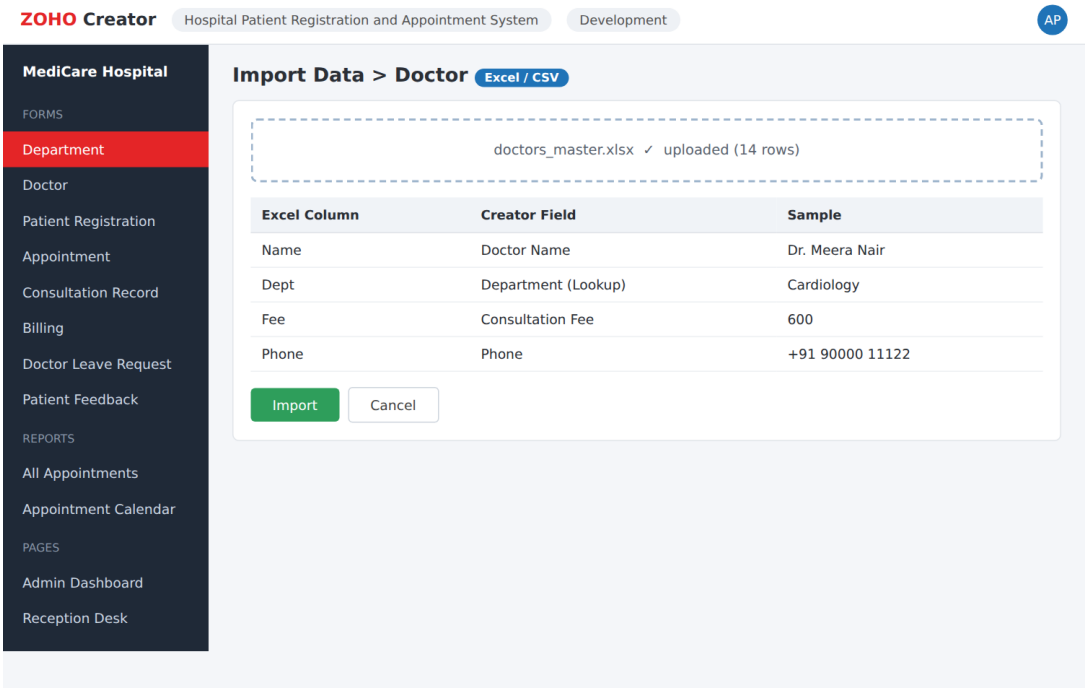


Figure 5.14: Data Import Setup

5.9 AI-Assisted Features

The Zia AI application builder was given a short text prompt describing the project. It suggested five forms (Patient, Doctor, Department, Appointment and Billing). These were reviewed and rebuilt manually with the field types, lookups, rules and validations of Chapter 4, and the Consultation, Leave Request and Feedback forms, all reports, pages and workflows were created by the author without AI. Figure 5.15 shows the AI builder.

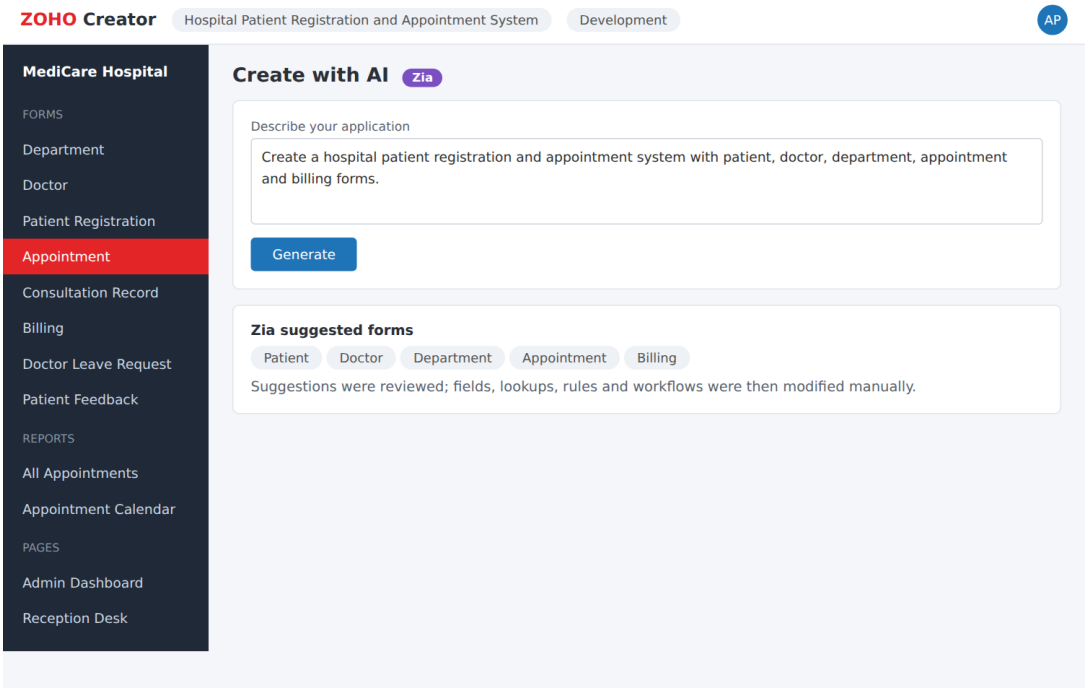


Figure 5.15: AI Application Builder in Use

5.10 Mobile Application

The application was opened on the Zoho Creator mobile app. The Patient Registration form uses the phone camera to capture the patient photo, and the Appointments report shows cards with status colours. Push notifications from the confirmation and reminder workflows are received on the phone (Figure 5.16).

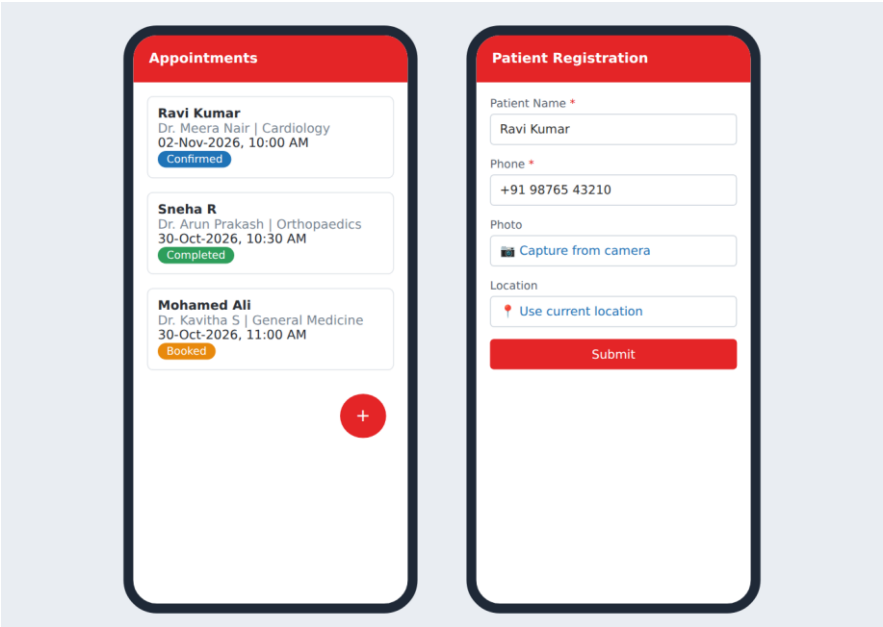


Figure 5.16: Application on the Mobile App

CHAPTER 6
TESTING

6.1 Testing Approach

Testing was carried out in the Development environment with sample records. Each form was tested with correct and wrong inputs, including empty mandatory fields, duplicate phone numbers and invalid dates. Each workflow was triggered by adding, editing or scheduling a record and its actions were verified in the notification log. The reports and dashboard were checked against a known set of 40 sample appointments, and each role was tested by logging in with the demo accounts to confirm menus and record-level permissions. Test records were deleted before publishing.

6.2 Test Cases

Eighteen test cases covering every functional requirement of Table 3.2, including negative tests, are given in Table 6.1.

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Register a patient with all mandatory fields	Fill all fields and submit	Record saved; UHID generated; success message	As expected	Pass
TC-02	FR-01	Submit registration with Phone empty	Leave Phone blank and submit	Error “Phone is required”; not saved	As expected	Pass
TC-03	FR-01	Register a patient with an existing phone number	Enter an existing phone	Error for unique field; not saved	As expected	Pass
TC-04	FR-02	Age calculation	Enter DOB 14-Mar-1988	Age shows 38 (as of Oct 2026)	As expected	Pass
TC-05	FR-03	Book an appointment in a free slot	Select patient, doctor, date, slot	Appointment saved with APT- ID	As expected	Pass
TC-06	FR-03	Book an already booked slot	Same doctor, date and slot as an existing one	Alert “Slot already booked”; not saved	As expected	Pass
TC-07	FR-03	Book an appointment for a past date	Select yesterday’s date	Validation error shown	As expected	Pass
TC-08	FR-04	Confirmation message	Book a valid appointment	Email and notification received; status Confirmed	As expected	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-09	FR-05	Reminder	Appointment on next day with status Confirmed	Reminder email received at 9:00 AM	As expected	Pass
TC-10	FR-07	Consultation completion	Save a Consultation Record	Appointment status = Completed; bill created	As expected	Pass
TC-11	FR-08	Net amount calculation	Fee 600, charges 200, discount 5%	Discount 40; Net Amount 760	As expected	Pass
TC-12	FR-08	Discount above 10%	Enter discount 15%	Sent for approval to Administrator	As expected	Pass
TC-13	FR-09	Doctor leave approval	Submit leave request	Pending with Administrator; status updated on decision	As expected	Pass
TC-14	FR-10	No-show marking	Leave a Confirmed appointment unattended	Status = No-Show at 8:00 PM	As expected	Pass
TC-15	FR-11	Dashboard access by Administrator	Open Admin Dashboard	KPIs and charts displayed	As expected	Pass
TC-16	NFR-02	Receptionist opens Admin Dashboard	Login as Receptionist	Page not visible; access denied	As expected	Pass
TC-17	FR-13	Patient views other patient's appointment	Login as Patient; open My Appointments	Only own appointments displayed	As expected	Pass
TC-18	FR-15	Import doctors from Excel	Upload doctors_master.xlsx	14 records imported	As expected	Pass

6.3 Problems Faced and Fixes

Table 6.2: Issue Log

Issue ID	Problem Observed	Cause	Fix Applied	Status
ISS-01	Doctor drop-down showed doctors of all departments	Lookup filter not set	Added a lookup filter: Doctor.Department = selected Department	Closed
ISS-02	Two patients could be given the same slot	No composite unique check available	Added the Slot Clash Check validation workflow	Closed
ISS-03	Reminder email sent for cancelled appointments	Schedule had no criteria	Added criteria Status = Confirmed	Closed
ISS-04	Receptionist could see the Consultation Record	Permission set copied from Administrator	Removed the form from the Receptionist permission set	Closed

Issue ID	Problem Observed	Cause	Fix Applied	Status
ISS-05	Net Amount showed a decimal error when Discount % was empty	Empty value in formula	Set default value 0 for Discount % and Other Charges	Closed

6.4 User Acceptance Testing

Three people used the application in assigned roles and gave the feedback of Table 6.3.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Classmate (Mechanical Engineering, Semester V)	Receptionist	Booking took less than two minutes; wanted the Doctor list to show the department name.	Department added as a display column in the Doctor lookup.
Lab assistant, Mechanical Engineering Dept.	Administrator	Dashboard is clear; asked for a pending-bills figure.	Pending Bills KPI added to the dashboard.
Family member (staff nurse)	Patient / Doctor	Mobile screens are easy to use; wanted the status colours.	Status colours enabled in the mobile list view.

CHAPTER 7
PUBLISHING, SHARING AND SECURITY

7.1 Publishing the Application

After testing, the application was published from the Development to the Production environment. The Administrator, a Receptionist, a Doctor and a Patient demo user were invited through **Users & Control → Users** and assigned to their roles. The Book Appointment form was shared through the customer portal for registered patients, the public permalink was kept disabled, and the Zoho Creator mobile app was verified with the same logins. Figure 7.1 shows the publish and share settings, and Table 7.1 is the publishing checklist.

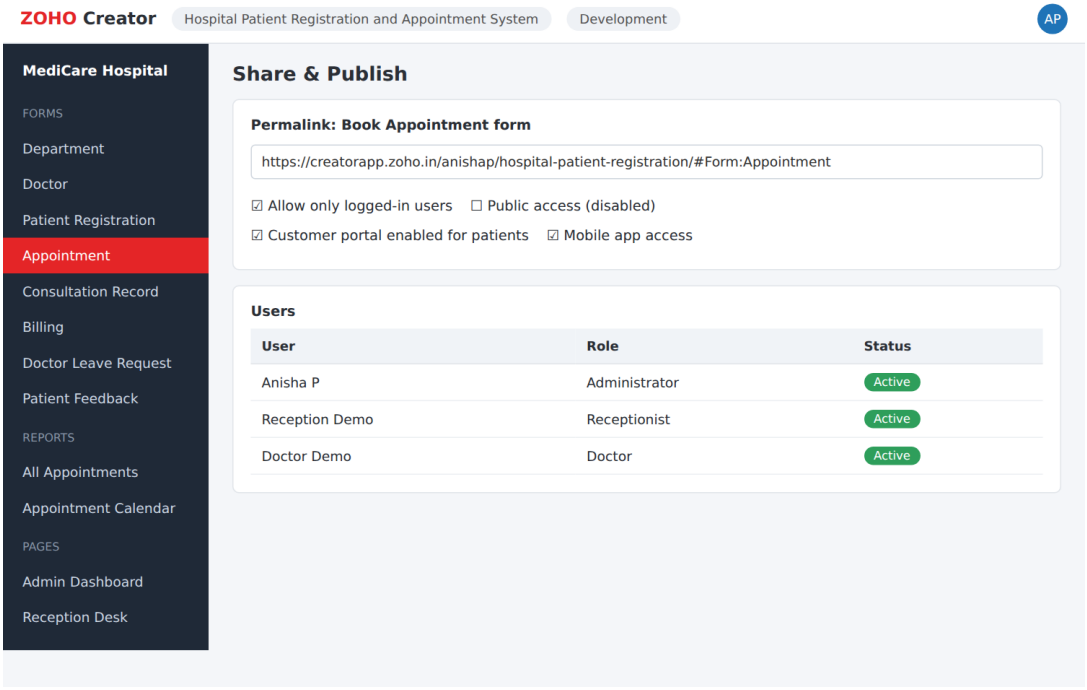


Figure 7.1: Publish / Share Settings

Table 7.1: Publishing Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed and passed	✓
2	Test records removed	✓
3	Users invited and roles assigned	✓
4	Workflows and schedules switched on	✓
5	Application opened and checked on the mobile app	✓
6	Application link verified by another user	✓

7.2 Security and Data Protection

Patient data is sensitive. The security settings of Table 7.2 were applied so that each user sees only what the role requires. Zoho Creator also encrypts data in transit and at rest and keeps an audit trail of record changes. Because the application uses sample data only, no real patient information has been entered.

Table 7.2: Security Settings Applied

Security Setting	Where Applied	Risk Prevented
Record-level rule: own records only	Doctor and Patient roles	Users seeing other patients' data
Consultation form hidden from Receptionist	Permission set	Exposure of clinical details
Billing hidden from Doctor role	Permission set	Misuse of financial data
Published public link disabled	Share settings	Anonymous public access
Two-factor authentication for Administrator	Zoho account	Account takeover
Mandatory consent checkbox	Patient Registration	Treatment records without consent

7.3 Administration and Maintenance

The author is the application Administrator. Changes are made first in the Development environment and then published to Production. Data can be exported to Excel at any time and a backup of the application can be taken from the settings. The audit history of every record shows who added or edited it, and the workflow log is used to check that notifications and schedules have run. New doctors and departments are added through the forms, and a new role can be added from Users & Control without disturbing existing data.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

One realistic scenario is followed through the roles. A patient, Ravi Kumar (UHID-0249), calls the hospital for a cardiology consultation. In Step 1 the receptionist books an appointment with Dr. Meera Nair for 2 November 2026 at 10:00 AM; the slot check passes and a confirmation email is sent. In Step 2 a doctor's leave request submitted for another doctor is approved by the Administrator through the approval process. In Step 3 the consultation is recorded with a two-row prescription, the appointment status becomes Completed and a bill is generated automatically. The Admin Dashboard then updates its counts.

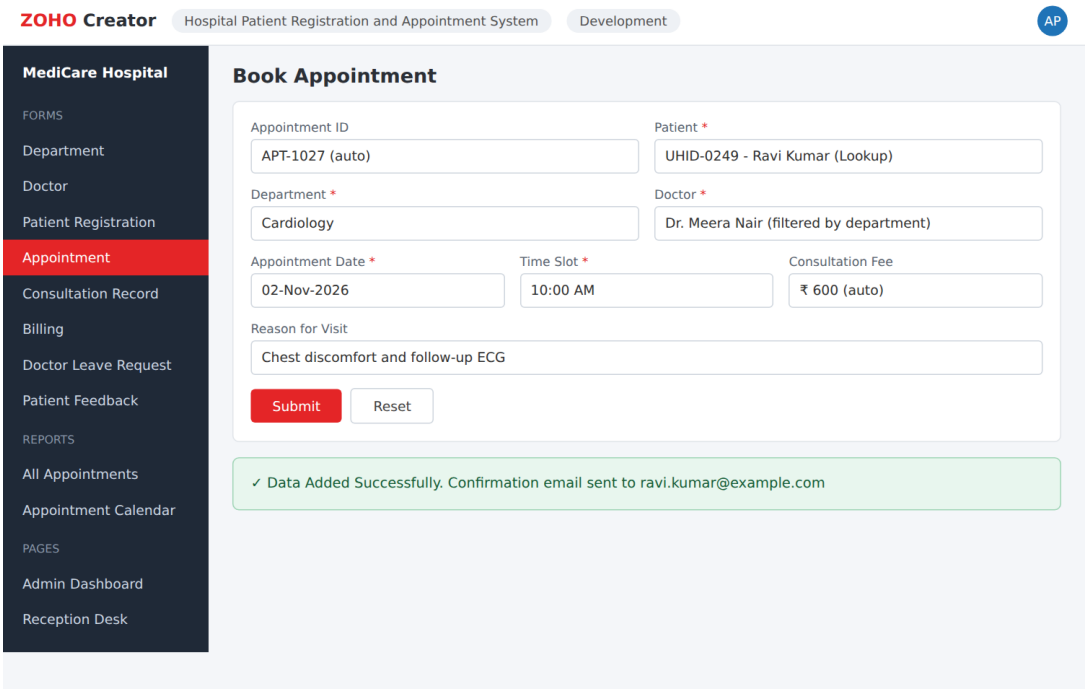


Figure 8.1: Scenario Step 1 – Receptionist Books the Appointment

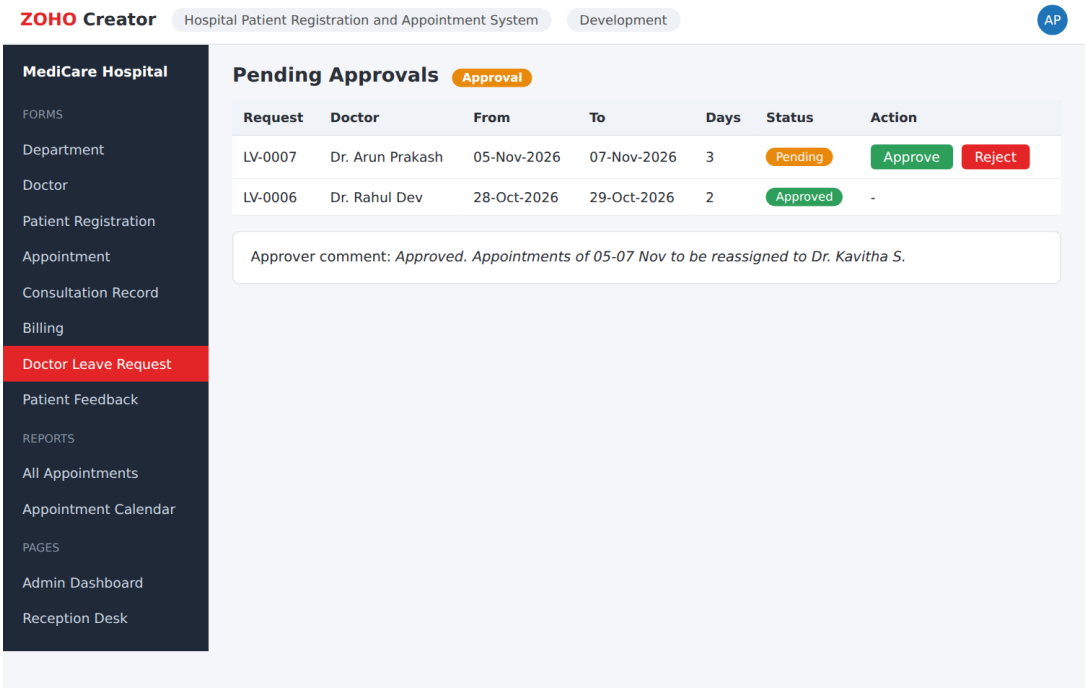


Figure 8.2: Scenario Step 2 – Administrator Approves a Doctor Leave Request

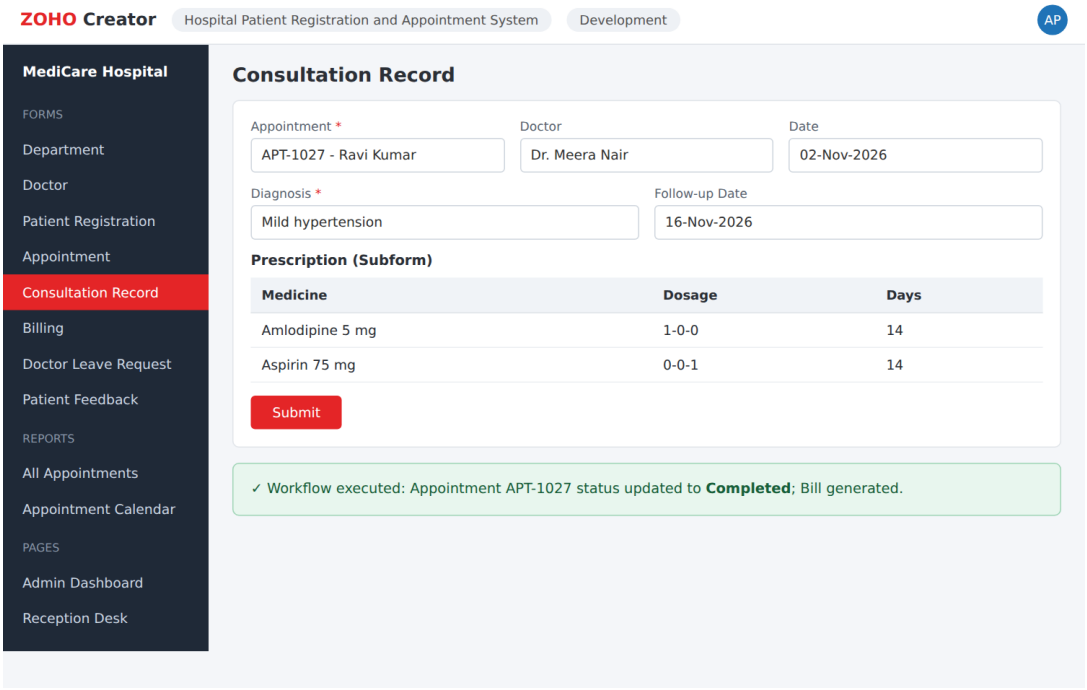


Figure 8.3: Scenario Step 3 – Doctor Records the Consultation

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Achieved Through	Outcome / Evidence
Digitise records using forms with validations	Section 5.2, 5.3; Tables 4.2 – 4.9	Eight forms, 24 field types, validations working (TC-01 to TC-04)

Objective (Sec. 1.3)	Achieved Through	Outcome / Evidence
Provide reports and a dashboard	Sections 5.4, 5.5	Ten reports and two pages; dashboard in Figure 5.9
Automate notifications, approvals and updates	Section 5.6; Appendix C	Eight workflows (4 normal, 2 approval, 2 schedule); TC-05 to TC-14
Restrict access by role	Sections 4.6, 5.7, 7.2	Four roles with record-level rules; TC-16, TC-17
Publish for web and mobile use	Sections 5.10, 7.1	Published to four demo users, mobile app verified

8.3 Comparison with the Existing System

The values below were estimated from trial runs with sample data and a manual walk-through of the paper process; they indicate the order of improvement and are not clinical measurements.

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Time to register and book one patient	About 10 – 12 minutes	About 2 – 3 minutes	About 75% faster
Paper / manual steps	6 manual steps (form, register, file card, diary, bill, summary)	0 paper steps for registration and booking	Paperless
Time to prepare daily summary report	About 45 minutes	Instant (dashboard)	About 100% saved
Missed / delayed tasks	Frequent (no reminders, double booking)	Reduced by reminders, slot check and approvals	Significant reduction

8.4 Limitations of the No-Code Approach

Some needs could not be met by no-code tools alone. A composite unique check (doctor, date and slot together) is not available as a field property and had to be handled through a validation workflow. Fully customised screen layouts, an online payment gateway, SMS delivery through a hospital's own sender ID, and integration with laboratory machines or biomedical devices would require low-code scripting or external integrations. Complex calculations such as dynamic slot generation based on average consultation time are also difficult. A low-code or pro-code extension using Deluge scripts or REST APIs can be added to the same application when these needs arise.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Hospital Patient Registration and Appointment System was designed, built, tested and published on Zoho Creator without writing any code. It contains eight forms with 24 field types, ten reports, two pages and eight workflows of normal, approval and schedule types, supported by four user roles. It replaces the paper register with a single patient identity, prevents double booking, reminds patients, automates billing and approvals, and gives the administrator a live dashboard.

Testing with eighteen test cases and three users showed that all functional requirements were met. The project shows that a non-software engineer can deliver a working business application in a short time using a no-code platform, while also revealing the limits of such platforms that call for low-code extensions.

9.2 Future Enhancements

- Add a QR code to each patient card so that the front desk can open the patient record by scanning it.
- Use Zia prediction to forecast daily appointment load and no-show probability.
- Integrate an online payment gateway and Zoho Analytics for deeper reporting using low-code scripting.
- Add a Blueprint for the in-patient admission process and a telemedicine video-consultation link.

9.3 Relevance to My Discipline

Mechanical engineers are regularly asked to plan maintenance schedules, track work orders, manage service centres and report key performance indicators. The same patterns appear in this project: a limited resource (the doctor) is scheduled against demand, requests pass through approvals, and a dashboard summarises performance. Building the application has given me skills in data modelling, process mapping, role-based security and workflow automation, and has shown how a citizen developer or digital-transformation engineer can digitise workshop, plant and service operations without depending on a software team.

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APPENDIX A – ZOHO CREATOR NO-CODE FEATURE CHECKLIST

The Unit column follows the five units of the course (I – Introduction to no-code and setup, II – Forms and logic, III – Reports and pages, IV – Workflows and automation, V – Users, data, publishing and testing).

Table A.1: No-Code Feature Coverage Checklist

Area	Zoho Creator Feature	Done	Unit	Where in the Application	Section / Figure
Setup	Application created; sections/menu, theme and icon configured	Y	I	Application home and menu	5.1 / Fig. 5.1
Forms	Forms using ≥ 10 distinct field types	Y	II	All 8 forms (24 field types)	4.3 / Fig. 5.2
Forms	Lookup field linking two forms	Y	II	Appointment → Patient, Doctor	4.2 / Table 4.1
Forms	Subform (multiple rows in one record)	Y	II	Consultation Record → Prescription	4.3 / Fig. 8.3
Forms	Mandatory, unique and default-value properties	Y	II	Patient, Doctor, Appointment	5.2 / Fig. 5.3
Forms	Media fields (Image / File / Signature / Barcode-QR)	Y	II	Photo, Lab Report, Doctor Signature	4.3 / Fig. 5.2
Logic	Formula field	Y	II	Age, Total Days, Net Amount	5.3 / Fig. 5.4
Logic	Field rules / conditional show–hide	Y	II	Appointment, Leave Request	5.3 / Fig. 5.4
Logic	Validation rule with error message	Y	II	Appointment Date, Leave Dates	5.3 / Fig. 5.4
Reports	List report with filter, grouping and sorting	Y	III	All Appointments	5.4 / Fig. 5.5
Reports	Kanban / Calendar / Timeline / Map report	Y	III	Appointment Board, Calendar	5.4 / Fig. 5.6, 5.7
Reports	Pivot table or chart report	Y	III	Revenue Summary, Dept. chart	5.4 / Fig. 5.8
Reports	Conditional formatting or record template	Y	III	All Appointments, Patient Directory	5.4 / Fig. 5.5
Pages	Dashboard page with KPI widgets and charts	Y	III	Admin Dashboard	5.5 / Fig. 5.9
Pages	Page with embedded form / report and buttons	Y	III	Reception Desk	5.5
Workflow	Form workflow with email / notification action	Y	IV	Appointment Confirmation	5.6 / Fig. 5.10
Workflow	Form workflow updating a field or creating a record	Y	IV	Consultation Completed	5.6 / Table 5.3
Workflow	Scheduled / reminder workflow	Y	IV	Appointment Reminder	5.6 / Fig. 5.11

Area	Zoho Creator Feature	Done	Unit	Where in the Application	Section / Figure
Workflow	Approval process	Y	IV	Doctor Leave, Billing Discount	5.6 / Fig. 5.12
Workflow	Blueprint (stage-based process)	N	IV	Not used (listed as future work)	9.2
Users	Users invited; roles and permissions configured	Y	V	Four roles	5.7 / Fig. 5.13
Users	Record-level access rule	Y	V	Doctor and Patient roles	5.7
Data	Import from Excel / CSV	Y	V	Doctor master	5.8 / Fig. 5.14
Data	Export / share report (PDF, Excel, link)	Y	V	Doctor Directory	5.8
Data	Built-in integration with another Zoho app	Y	V	Zoho Mail notifications	5.8
AI	AI app builder / Zia feature used	Y	V	Zia app builder	5.9 / Fig. 5.15
Publish	Application published to users / portal	Y	V	Patient portal	7.1 / Fig. 7.1
Publish	Used on the Zoho Creator mobile app	Y	V	Registration and appointments	5.10 / Fig. 5.16
Security	Security settings applied (hidden fields, link control, 2FA)	Y	V	Permission sets, share settings	7.2 / Table 7.2
Testing	Test cases and user acceptance testing documented	Y	V	Chapter 6	6.2 – 6.4

Total features configured: 29 / 30

APPENDIX B – COURSE OUTCOME MAPPING

The Course Outcomes are stated in the manner of the OCB2331 syllabus; the examiner may revise the levels.

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the concepts of no-code and low-code development and set up an application on Zoho Creator.	Ch. 1, 2, 5.1	3	
CO2	Design forms with suitable field types, validations, lookups, subforms and rules.	Ch. 4, 5.2, 5.3	3	
CO3	Create reports, pages and dashboards for data presentation.	Ch. 5.4, 5.5	3	
CO4	Automate business processes using workflows, schedules and approvals.	Ch. 5.6, App. C	3	
CO5	Manage users, roles, security, data, publishing and testing of the application.	Ch. 5.7 – 5.10, 6, 7	2	

APPENDIX C – APPLICATION CONFIGURATION RECORD

Table C.1: Summary of Application Components

Component Type	Name	Purpose	Section Ref.
Form	Department	Master list of hospital departments	4.3, 5.2
Form	Doctor	Master list of consultant doctors	4.3, 5.2
Form	Patient Registration	Master record of each registered patient	4.3, 5.2
Form	Appointment	Each OPD appointment booked for a patient	4.3, 5.2
Form	Consultation Record	Clinical record of a completed consultation	4.3, 5.2
Form	Billing	Bill raised for each appointment	4.3, 5.2
Form	Doctor Leave Request	Leave applications of doctors	4.3, 5.2
Form	Patient Feedback	Feedback collected after consultation	4.3, 5.2
Report	All Appointments	List report on Appointment	5.4
Report	Patient Directory	List report on Patient Registration	5.4
Report	Doctor Directory	List report on Doctor	5.4
Report	Appointment Board	Kanban report on Appointment	5.4
Report	Appointment Calendar	Calendar report on Appointment	5.4
Report	Appointments by Department	Chart (Bar) report on Appointment	5.4
Report	Revenue Summary	Pivot report on Billing	5.4
Report	Billing Sheet	Spreadsheet report on Billing	5.4
Report	Pending Leave Requests	List report on Doctor Leave Request	5.4
Report	My Appointments	List report on Appointment	5.4
Page	Admin Dashboard	KPIs and charts for the Administrator	5.5
Page	Reception Desk	Front-desk screen with forms and today’s list	5.5
Workflow	Patient Welcome Alert	Form workflow (Normal)	5.6
Workflow	Slot Clash Check	Form workflow (Normal)	5.6
Workflow	Appointment Confirmation	Form workflow (Normal)	5.6
Workflow	Consultation Completed	Form workflow (Normal)	5.6
Workflow	Doctor Leave Approval	Approval	5.6
Workflow	Billing Discount Approval	Approval	5.6
Workflow	Appointment Reminder	Schedule (date-based)	5.6
Workflow	Daily No-Show Marker	Schedule (recurring)	5.6
Role	Administrator	Full access and approvals	5.7
Role	Receptionist	Registration, booking and billing	5.7
Role	Doctor	Own appointments and consultations	5.7
Role	Patient	Portal: own appointments and feedback	5.7

Table C.2: Workflow Settings in Detail – Patient Welcome Alert

Form / Report	Patient Registration
Run when	Record created (On Add → On Success)
Condition	Email is not empty
Action 1	Send email → To: Email | Subject: Welcome to MediCare Hospital, your UHID is <Patient ID>
Action 2	Send notification → To: Receptionist | Message: New patient registered
Action 3	—

Table C.3: Workflow Settings in Detail – Slot Clash Check

Form / Report	Appointment
Run when	Record created or edited (On Validate)
Condition	Count of Appointment with same Doctor, Appointment Date and Time Slot, Status ≠ Cancelled, > 0
Action 1	Show alert → "This slot is already booked. Please choose another slot."
Action 2	Stop form submission
Action 3	—

Table C.4: Workflow Settings in Detail – Appointment Confirmation

Form / Report	Appointment
Run when	Record created (On Add → On Success)
Condition	Status = Booked
Action 1	Send email → To: Patient Email, Doctor Email | Subject: Appointment confirmed
Action 2	Send SMS / push notification → To: Patient Phone
Action 3	Update field → Status = Confirmed

Table C.5: Workflow Settings in Detail – Consultation Completed

Form / Report	Consultation Record
Run when	Record created (On Add → On Success)
Condition	Appointment is not empty
Action 1	Update field → Appointment.Status = Completed
Action 2	Create record → Billing (Appointment, Consultation Fee)
Action 3	Send email → Feedback request to patient

Table C.6: Workflow Settings in Detail – Doctor Leave Approval

Form / Report	Doctor Leave Request
Run when	Record created (approval starts on add)
Condition	Total Days ≥ 1

Action 1	Approver → Administrator (single level, reminder after 24 hours)
Action 2	On Approve → Approval Status = Approved; email doctor
Action 3	On Reject → Approval Status = Rejected; email with comments

Table C.7: Workflow Settings in Detail – Billing Discount Approval

Form / Report	Billing
Run when	Record created or edited
Condition	Discount % > 10
Action 1	Approver → Administrator (single level)
Action 2	On Approve → Approval Status = Approved
Action 3	On Reject → Discount % = 0; Approval Status = Rejected

Table C.8: Workflow Settings in Detail – Appointment Reminder

Form / Report	Appointment
Run when	Date-based schedule: 1 day before Appointment Date at 9:00 AM
Condition	Status = Confirmed
Action 1	Send email → To: Patient | Subject: Appointment reminder
Action 2	Send push notification → To: Patient
Action 3	—

Table C.9: Workflow Settings in Detail – Daily No-Show Marker

Form / Report	Appointment
Run when	Recurring schedule: daily at 8:00 PM
Condition	Appointment Date = today and Status in (Booked, Confirmed)
Action 1	Update field → Status = No-Show
Action 2	Send email → To: Administrator | Subject: Daily no-show summary
Action 3	—

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the application link in a browser (Chrome, Edge or Firefox), enter the registered email and password and click Sign In. On a phone, install the Zoho Creator app, sign in with the same account and open "Hospital Patient Registration and Appointment System". Patients use the portal link sent to them by email.

D.2 Instructions for Receptionist

1. Open the Reception Desk page.
2. To register a new patient, click Register Patient, fill the mandatory fields, tick the consent box and click Submit. Note the UHID shown.
3. To book an appointment, click Book Appointment, choose the patient, department, doctor, date and slot, and click Submit.
4. If "Slot already booked" appears, choose another slot.
5. Use the Appointment Board to move a booking between statuses, and the Billing Sheet to enter payment details.

D.3 Instructions for Doctor

1. Open the All Appointments report; only your appointments are shown.
2. After the visit, open Consultation Record, select the appointment, enter symptoms and diagnosis, and add the prescription rows.
3. Click Submit; the appointment is marked Completed automatically.
4. To apply for leave, open Doctor Leave Request, fill the dates and submit.

D.4 Instructions for Administrator

1. Open the Admin Dashboard to view patients, appointments, bills and cancellations.
2. Open Pending Leave Requests and click Approve or Reject; add a comment.
3. Approve or reject high-discount bills from the approval notification.
4. Add departments, doctors and users from the forms and Users & Control.

D.5 Instructions for Patient

1. Sign in to the patient portal.
2. Open Book Appointment, choose the department, doctor, date and slot, and submit.
3. View or cancel your bookings in My Appointments.
4. After the visit, open Patient Feedback and rate the visit.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	ANISHA P	Register No.	241101009
Branch	Mechanical Engineering	Date	31 October 2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Relevance to own discipline, clear objectives, complete FR/NFR, use cases	10	
2	Application Design	Data model, form specifications, process flows, permission matrix	10	
3	Forms, Formula Fields & Rules	Variety of field types, lookups/subforms, formulas, field and validation rules	15	
4	Reports, Pages & Dashboards	Variety of report types, filters/formatting, dashboard design	15	
5	Workflows & Automation	Form workflows, schedules, approvals, Blueprint configured without code	15	
6	Users, Security, Data & Publishing	Roles and permissions, record rules, import/export, publishing, mobile app	10	
7	Live Demonstration	End-to-end working across roles; feature checklist verified	10	
8	Project Report	Adherence to template and format, clarity, screenshots, references	5	
9	Viva-Voce	Understanding of the platform, design choices and no-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90 – 100	Feature configured completely and correctly, beyond the basic requirement; student explains choices confidently.
Good	70 – 89	Feature configured correctly with minor gaps; explanation adequate.
Satisfactory	50 – 69	Feature partially configured or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, not working, or not understood by the student.

Signature of Internal Examiner Name: _____	**Signature of External Examiner** Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	ANISHA P	Register No.	241101009
Branch / Section	Mechanical Engineering	Date of Viva	31 October 2026
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1 – 8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

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_____

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INTERNAL EXAMINER Name: _____	**EXTERNAL EXAMINER** Name: _____